

C2 Assignment : Differential Calculus and Trigonometry

SINGLE CORRECT TYPE QUESTIONS

D05/1/001. Let $f(x) = (x^2 - 3x + 2) |(x^3 - 6x^2 + 11x - 6)| + \left| \sin \left(x + \frac{\pi}{4} \right) \right|$. Number of points at which the function $f(x)$ is non-differentiable in $[0, 2\pi]$, is :

माना $f(x) = (x^2 - 3x + 2) |(x^3 - 6x^2 + 11x - 6)| + \left| \sin \left(x + \frac{\pi}{4} \right) \right|$ है। उन बिन्दुओं की संख्या जिन पर $f(x)$, $[0, 2\pi]$ में

अवकलनीय नहीं है, होगी :

- (A) 5 (B) 4 (C) 3 (D) 2

Ans. C

Sol. We have $f(x) = (x - 1)(x - 2) |(x - 1)(x - 2)(x - 3)| + \left| \sin \left(x + \frac{\pi}{4} \right) \right|$

Clearly, $f(x)$ is not differentiable at $x = 3, \frac{3\pi}{4}, \frac{7\pi}{4}$

D05/1/002. If n_1 = number of points of non derivability of $f(x) = \begin{cases} \max. \{|x|, \sin|x|\}, & x \in \left(0, \frac{\pi}{2}\right) \\ \max. \{|x|, \tan|x|\}, & x \in \left(\frac{-\pi}{2}, 0\right) \end{cases}$ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and

n_2 = number of points of discontinuity of $g(x) = [\sin x] + [\cos x]$ in $[0, 2\pi]$ then $(n_1 + n_2)$ is equal to :

यदि $n_1, f(x) = \begin{cases} \max. \{|x|, \sin|x|\}, & x \in \left(0, \frac{\pi}{2}\right) \\ \max. \{|x|, \tan|x|\}, & x \in \left(\frac{-\pi}{2}, 0\right) \end{cases}$ के $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में उन बिन्दुओं की संख्या है जिन पर फलन

अवकलनीय नहीं है तथा $n_2, g(x) = [\sin x] + [\cos x]$ के $[0, 2\pi]$ में असतता के बिन्दुओं की संख्या है तब $(n_1 + n_2)$ का मान बराबर होगा :

- (A) 3 (B) 4 (C) 5 (D) 6

Ans. D

Sol. Clearly, $f(x) = \begin{cases} |x| = x; & x \in \left(0, \frac{\pi}{2}\right) \\ \tan(-x); & x \in \left(\frac{-\pi}{2}, 0\right) \end{cases}$ so $n_1 = 1$

$$g(x) = \begin{cases} 1, & x = 0 \\ 0, & 0 < x < \frac{\pi}{2} \\ 1, & x = \frac{\pi}{2} \\ -1, & \frac{\pi}{2} < x \leq \pi \\ -2, & \pi < x < \frac{3\pi}{2} \\ -1, & \frac{3\pi}{2} \leq x < 2\pi \\ 1, & x = 2\pi \end{cases}$$

$$\Rightarrow n_2 = 5$$

$$\therefore n_1 + n_2 = 6$$

D05/1/003. Let $f(x)$, $g(x)$ be function on the real line R such that both $f(x) + g(x)$ and $f(x)g(x)$ are, differentiable. Which of the following is **FALSE**?

- (A) $(f(x))^2 + (g(x))^2$ is necessarily differentiable
- (B) $f(x)$ is differentiable if and only if $g(x)$ is differentiable
- (C) $f(x)$ and $g(x)$ are necessarily continuous
- (D) If $f(x) > g(x)$ for all $x \in R$, then $f(x)$ is differentiable

माना फलन $f(x)$, $g(x)$ वास्तविक रेखा R पर इस प्रकार हैं कि $f(x) + g(x)$ तथा $f(x)g(x)$ दोनों अवकलनीय हैं। निम्न में से कौनसा गलत है ?

- (A) $(f(x))^2 + (g(x))^2$ आवश्यक रूप से अवकलनीय है
- (B) $f(x)$ अवकलनीय है यदि और यदि $g(x)$ अवकलनीय है
- (C) $f(x)$ तथा $g(x)$ आवश्यक रूप से सतत हैं
- (D) यदि $f(x) > g(x)$ सभी $x \in R$ के लिए, तब $f(x)$ अवकलनीय है

Ans. C

Sol. Let $f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$, $g(x) = \begin{cases} 0, & x \geq 0 \\ 1, & x < 0 \end{cases}$

Then $f(x) + g(x) = 1$, $f(x)g(x) = 0$ are differentiable, while both $f(x)$ and $g(x)$ are discontinuous.

D05/1/004. Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} (1 - \cos x) \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Then, :

- (A) f is discontinuous
- (B) f is continuous but not differentiable
- (C) f is differentiable and its derivative is discontinuous
- (D) f is differentiable and its derivative is continuous

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \begin{cases} (1 - \cos x) \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

तब :

- (A) f असतत् है
- (B) f सतत् है लेकिन अवकलनीय नहीं
- (C) f अवकलनीय है तथा $f'(x)$ असतत् है
- (D) f अवकलनीय है तथा $f'(x)$ सतत् है

Ans. C

Sol. We compute the derivative of f . If $x \neq 0$, then

$$f'(x) = \sin x \sin \frac{1}{x} - \frac{1 - \cos x}{x^2} \cos \frac{1}{x}$$

If $x = 0$, then

$$f'(0) = \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \sin \frac{1}{x} = 0$$

$$\text{as } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0 \text{ and } \left| \sin \frac{1}{x} \right| \leq 1. \text{ So,}$$

$$f'(x) = \begin{cases} \sin x \sin \frac{1}{x} - \frac{1 - \cos x}{x^2} \cos \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

The derivative $f'(x)$ is not continuous at $x = 0$, as for $x_n = \frac{1}{2\pi n}$ we have

$$f'(x_n) = -\frac{1 - \cos x_n}{x_n^2} \rightarrow -\frac{1}{2}$$

D05/1/005. Consider the following functions

$$f(x) = \begin{cases} 1, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases} \text{ and } f(g) = \begin{cases} 1, & |x| \leq 2 \\ 2, & |x| > 2 \end{cases}$$

Define $h_1(x) = f(g(x))$ and $h_2(x) = g(f(x))$. Which of the following statements is correct :

- (A) h_1 and h_2 are continuous everywhere
- (B) h_1 is continuous everywhere and h_2 has discontinuity at ± 1
- (C) h_2 is continuous everywhere and h_1 has discontinuity at ± 2
- (D) h_1 has discontinuity at ± 2 and h_2 has discontinuity at ± 1

निम्न फलनों पर विचार कीजिए

$$f(x) = \begin{cases} 1, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases} \text{ तथा } f(g) = \begin{cases} 1, & |x| \leq 2 \\ 2, & |x| > 2 \end{cases}$$

$h_1(x) = f(g(x))$ तथा $h_2(x) = g(f(x))$ को परिभाषित कीजिए। निम्न में से कौनसा कथन सही होगा :

- (A) h_1 तथा h_2 सभी जगह सतत् हैं
- (B) h_1 सभी जगह सतत् है तथा h_2 , ± 1 पर असतत् है
- (C) h_2 सभी जगह सतत् है तथा h_1 , ± 2 पर असतत् है
- (D) h_1 , ± 2 पर असतत् है तथा h_2 , ± 1 पर असतत् है

Ans. C

Sol.

D05/1/006. Let f be a positive continuous function satisfying

$$f(x^3 - 4x) = \frac{2}{f(2^x + 7) - 1} \quad \forall x \in \mathbb{R}$$

Also, let g be a continuous function satisfying

$$g(x) = g(2x) \quad \forall x \in \mathbb{R}, \quad g(1) = \frac{\pi}{2},$$

then find

$$\lambda = f(15) + \lim_{x \rightarrow 0} \frac{\cos^2 g(x) + 1 - \sin^3 g(x)}{\sin^2 x}.$$

माना $f, f(x^3 - 4x) = \frac{2}{f(2^x + 7) - 1} \forall x \in \mathbb{R}$ को संतुष्ट करने वाला एक धनात्मक सतत फलन है।

माना $g, g(x) = g(2x) \forall x \in \mathbb{R}, g(1) = \frac{\pi}{2}$ को संतुष्ट करने वाला एक सतत फलन है।

तब $\lambda = f(15) + \lim_{x \rightarrow 0} \frac{\cos^2 g(x) + 1 - \sin^3 g(x)}{\sin^2 x}$ का मान ज्ञात कीजिए

(A) 1

(B) 2

(C) 3

(D) 4

Ans.

B

Sol.

D05/1/007. If $f(x) = \begin{cases} \ln(1+|x|+(b-2)|x+1|) + a \tan^{-1} x + 2 & , -2 < x < 0 \\ b & , x = 0 \\ \frac{(c+2) \sin^{-1} \sqrt{\{x\}} + e^{2x} - e^{\ln\left(4\left\{\frac{x}{2}\right\}\right)} - 1}{x^2} & , 0 < x < 2 \end{cases}$ is derivable in $(-2,2)$ then find the

value of $(9a^2 + b^2 + c^2)$

[Note : $\{y\}$ denotes fractional part of y]

यदि $f(x) = \begin{cases} \ln(1+|x|+(b-2)|x+1|) + a \tan^{-1} x + 2 & , -2 < x < 0 \\ b & , x = 0 \\ \frac{(c+2) \sin^{-1} \sqrt{\{x\}} + e^{2x} - e^{\ln\left(4\left\{\frac{x}{2}\right\}\right)} - 1}{x^2} & , 0 < x < 2 \end{cases}$ $(-2,2)$ में अवकलनीय है तब $(9a^2 + b^2$

+ c²) का मान ज्ञात कीजिए।

[नोट : {y}, y के भिन्नात्मक भाग को दर्शाता है]

(A) 60

(B) 40

(C) 57

(D) 58

Ans.

C

Sol.

D05/1/008. Let f_1 be a differentiable function satisfying

$$f_2(x) = \lim_{n \rightarrow \infty} \binom{n}{k} \left(f_1\left(\frac{x}{n}\right) - f_1(0) \right)^k;$$

$$f_1'(0) = \frac{1}{3},$$

$$f_3(k) = \lim_{x \rightarrow 0} \frac{1}{x} \sum_{r=1}^k f_3\left(\frac{x}{r}\right);$$

$$k \in \mathbb{N}, f_3(0) = 0, f_3'(0) = 1.$$

Then which of the following option(s) is/are equal to :

$$\lambda = \sum_{r=0}^{10} \left(\binom{10}{r} r! \cdot f_2(3) + \frac{f_3(r+1) - f_3(r)}{f_3(11)} \right)$$

माना f_1 एक अवकलनीय फलन है जो निम्न को संतुष्ट करता है :

$$f_2(x) = \lim_{n \rightarrow \infty} \binom{n}{k} \left(f_1\left(\frac{x}{n}\right) - f_1(0) \right)^k;$$

$$f_1'(0) = \frac{1}{3},$$

$$f_3(k) = \lim_{x \rightarrow 0} \frac{1}{x} \sum_{r=1}^k f_3\left(\frac{x}{r}\right);$$

$$k \in \mathbb{N}, f_3(0) = 0, f_3'(0) = 1.$$

$$\text{तब } \lambda = \sum_{r=0}^{10} \left(\binom{10}{r} r! \cdot f_2(3) + \frac{f_3(r+1) - f_3(r)}{f_3(11)} \right) \text{ का मान होगा}$$

(A) 1024

(B) 1026

(C) 1025

(D) 1203

Ans.

C

Sol.

D05/1/009. Let $\llbracket f(x) : A \rrbracket$ denote the number of points on non-differentiability of any function $y = f(x)$ in the set A. If

g is a polynomial function satisfying

$$g(x+y) = g(x) + g(y) + 2xy - 1 \quad \forall x \in \mathbb{R}, \quad f'(0) = 3,$$

then which of the following option(s) is/are equal to :

$$\lambda = \left\| \lim_{n \rightarrow \infty} \sqrt[n]{\sin^{2n} x + \cos^{2n} x} : [-2\pi, 2\pi] \right\| + \left\| |g(x) - 1| - 1 : \mathbb{R} \right\|$$

माना $\|f(x) : A\|$, $y = f(x)$ के किसी फलन के समुच्चय A में अन-अवकलनीयता के बिन्दुओं की संख्या को दर्शाता है। यदि g एक बहुपदीय फलन है जो $g(x+y) = g(x) + g(y) + 2xy - 1 \quad \forall x \in \mathbb{R}, f'(0) = 3$ को संतुष्ट करता है।

$$\text{तब } \lambda = \left\| \lim_{n \rightarrow \infty} \sqrt[n]{\sin^{2n} x + \cos^{2n} x} : [-2\pi, 2\pi] \right\| + \left\| |g(x) - 1| - 1 : \mathbb{R} \right\| \text{ का मान होगा :}$$

- (A) 14 (B) 15 (C) 16 (D) 17

Ans. A

Sol.

D02/1/010. Let $A = \{x_1, x_2, \dots, x_{50}\}$ and $B = \{y_1, y_2, \dots, y_{20}\}$ be two set of real numbers. What is the total number of function $f : A \rightarrow B$ such that f is nonto and $f(x_1) \leq f(x_2) \leq \dots \leq f(x_{50})$?

- (A) $\binom{49}{19}$ (B) $\binom{49}{20}$ (C) $\binom{50}{19}$ (D) $\binom{50}{20}$

Ans. A

Sol. To define an increasing mapping f of $A = \{x_1, x_2, \dots, x_{50}\}$ onto $B = \{y_1, y_2, \dots, y_{20}\}$ we must split elements of A into 20 nonempty groups of consecutive integers (first group will be mapped to y_1 , second – to y_2 and so on), i.e. to place 19 barriers on 49 available places between x'_j s.

D02/1/011. For a real number x , let $[x]$ denote the greatest integer less than or equal to x . Then the number of real solutions of $|2x - [x]| = 4$ is :

- (A) 4 (B) 3 (C) 2 (D) 1

Ans. A

Sol. $2x - [x] = \pm 4$. Let $x = n + r$, $n \in \mathbb{Z}$, $0 \leq r < 1$. Then $[x] = n$ and

$$2n + 2r - n = \pm 4,$$

$$n + 2r = \pm 4, 2r = \pm 4 - n.$$

Either $r = 0$ and $x = n = \pm 4$, or $r = \frac{1}{2}$, $n = 3$ or -5 , and $x = 3.5$ or -4.5 .

D02/1/012. Let p and q be two non-zero polynomials such that the degree of p is less than or equal to the degree of q , and $p(a)q(a) = 0$ for $a = 0, 1, 2, \dots, 10$. Which of the following must be :

- (A) degree of $q \neq 10$ (B) degree of $p \neq 10$ (C) degree of $q \neq 5$ (D) degree of $p \neq 5$

Ans. C

Sol. $p(a)q(a) = 0$, for $a = 0, 1, 2, \dots, 10$

Hence the polynomial, $P(x)q(x)$ has at least 11 roots as so $\deg(p(x)q(x)) \geq 11$

Now, if $\deg(q(x)) = 5$, then $\deg(p(x)) \leq 5 \Rightarrow \deg(p(x)q(x)) \leq 10$ which is a contradiction.

Degree $q \neq 5$

D02/1/013. Let $\{f_n(x)\}$ be a sequence of polynomials defined inductively as

$$f_1(x) = (x - 2)^2$$

$$f_{n+1}(x) = (f_n(x) - 2)^2, n \geq 1$$

Let a_n and b_n respectively denote the constant term and the coefficient of x in $f_n(x)$. Then :

(A) $a_n = 4, b_n = -4^n$

(B) $a_n = 4, b_n = -4n^2$

(C) $a_n = 4^{(n-1)!}, b_n = -4^n$

(D) $a_n = 4^{(n-1)!}, b_n = -4n^2$

Ans. A

Sol. $f_1(x) = (x - 2)^2; f_{n+1} = (f_n - 2)^2$ [Given]

$$f_1(x) = (x - 2)^2 = 4 - 4x + x^2$$

$$f_2(x) = (f_1(x) - 2)^2 = (2 - 4x + x^2)^2 = 2^2 + 2 \cdot 2(-4x) + \dots = 4 - 4^2x + \dots$$

$$f_3(x) = (f_2(x) - 2)^2 = (4 - 2^4x + \dots - 2)^2 = (2 - 2^4x + \dots)^2 \dots = 2^2 - 2 \cdot 2 \cdot 4^2x$$

$$f_4(x) = (f_3(x) - 2)^2 = 2^2 - 4^4x + \dots$$

$$f_5(x) = (f_4(x) - 2)^2 = 2^2 - 4^5x + \dots$$

$$\text{So, } a_n = 4, b_n = -4^{(\text{nth term of } 1, 2, 3, 4, 5)}$$

$\therefore$ A is the correct option.

D02/1/014. Let X be a nonempty set and let $P(X)$ denote the collection of all subsets of X . Define $f: X \times P(X) \rightarrow R$ by

$$f(x, A) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A. \end{cases} \text{ Then } f(x, A \cup B) :$$

(A) $f(x, A) + f(x, B)$

(B) $f(x, A) + f(x, B) - 1$

(C) $f(x, A) + f(x, B) - f(x, A) \cdot f(x, B)$

(D) $f(x, A) + |f(x, A) - f(x, B)|$

Ans. C

Sol. We will eliminate options.

(A) Assume $x \in A \cap B \Rightarrow x \in A$ and $x \in B$, Thus $f(x, A) + f(x, B) = 1 + 1 = 2$ which is greater than 1. Hence, incorrect.

(B) Assume $x \in A$ but $x \notin B$. This implies that $x \in A \cup B$. Thus, $f(x, A \cup B) = 1$ but $f(x, A) + f(x, B) - 1 = 1 + 0 - 1 = 0$. Hence, incorrect.

(C) Assume $x \in A \cap B \Rightarrow x \in A$ and $x \in B$. Therefore, $f(x, A) + f(x, B) - f(x, A) \cdot f(x, B) = 1 + 1 - 1 = 1$ Further, Assume $x \in A$ but $x \notin B$. This implies that $x \in A \cup B$. Therefore, $f(x, A) + f(x, B) - f(x, A) \cdot f(x, B) = 1 + 0 - 0 = 1$. Similarly, $x \in B$ but $x \notin A$. This option may be Correct.

(D) Assume $x \in A$ but $x \notin B$. This implies that $x \in A \cup B$. Thus, $f(x, A \cup B) = 1$ but $f(x, A) + |f(x, A) - f(x, B)| = 1 + |1 - 0| = 2$. Hence, incorrect.

∴ C is the correct answer.

D02/1/015. Let $S \subseteq \mathbb{R}$. Consider the statement :

"There exists a continuous function $f: S \rightarrow S$ such that $f(x) \neq x$ for all $x \in S$.

This statement is false if S equals

- (A) $[2, 3]$ (B) $(2, 3]$ (C) $[-3, -2] \cup [2, 3]$ (D) $(-\infty, \infty)$

Ans. A

Sol. (A) is false, i.e. $[2, 3]$ has the Fixed Point Property. Consider a continuous function

$$f: [2, 3] \rightarrow [2, 3]$$

and define

$$g: [2, 3] \rightarrow \mathbb{R}$$

$$g(x) = f(x) - x$$

It follows that $g(2) \geq 0$ and $g(3) \leq 0$. By the intermediate value theorem there is $x_0 \in [2, 3]$ such that $g(x_0) = 0$. Thus $f(x_0) - x_0 = 0$ and so $f(x_0) = x_0$.

(B) is true, $(2, 3]$ doesn't have the Fixed Point Property. For that take $f: (2, 3] \rightarrow (2, 3]$, $f(x) = \frac{1}{2}x + 1$.

(C) is true, $[-3, -2] \cup [2, 3]$ doesn't have the Fixed Point Property. For that take

$$f(x) = \begin{cases} 2 & \text{if } x \in [-3, -2] \\ -2 & \text{otherwise} \end{cases}$$

(D) is true, $(-\infty, \infty)$ doesn't have the Fixed Point Property : $f(x) = x + 1$.

D02/1/016. The number of polynomial functions f of degree ≥ 1 satisfying $f(x^2) = (f(x))^2 = f(f(x))$ for all real x , is :

- (A) 0 (B) 1 (C) 2 (D) infinitely many

Ans. B

Sol. Consider the equality $(f(x))^2 = f(f(x))$.

$$\text{Let } f(x) = a_0x^n + a_1x^{(n-1)} + a_2x^{(n-2)} + a_3x^{(n-3)} + \dots + a_n$$

Note that the degree of $(f(x))^2 = 2n$ and the degree of $f(f(x)) = n^2$

Since $(f(x))^2 = f(f(x))$, we have $2n = n^2$.

$$\Rightarrow n = 2.$$

$$\text{So, } f(x) = a_0x^2 + a_1x + a_2$$

Now, we have the equality $f(x^2) = (f(x))^2$

$$\text{Now, } f(x^2) = a_0x^4 + a_1x^2 + a_2 \text{ and}$$

$$(f(x))^2 = a_0^2x^4 + 2a_0a_1x^3 + (2a_0a_2 + a_1^2)x^2 + 2a_1a_2x + a_2^2$$

$$\text{Now we have, } a_0x^4 + a_1x^2 + a_2 = a_0^2x^4 + 2a_0a_1x^3 + (2a_0a_2 + a_1^2)x^2 + 2a_1a_2x + a_2^2 \rightarrow (1)$$

Now the coefficient of x^3 in L.H.S. is 0 and the coefficient of x^3 in R.H.S. is $2a_0a_1$.

$$\text{So, } 2a_0a_1 = 0.$$

Hence, $a_1 = 0$ (because a_0 can not be 0 since it is the leading coefficient)

Now we have by putting $a_1 = 0$ in equation (1)

$$a_0x^4 + a_2 = a_0^2x^4 + 2a_0a_2x^2 + a_2^2 \rightarrow (2)$$

Similarly, by comparing the coefficients of x^2 on both sides we get $a_2 = 0$.

So, equation (2) becomes

$$a_0 x^4 = a_0^2 x^4$$

$$\Rightarrow a_0 = a_0^2$$

$$\Rightarrow a_0 = 1.$$

So, the required polynomial function is $f(x) = x^2$

Hence, there is only one polynomial which x^2

So, correct option is B

D02/1/017. Given positive integer m , we define $f(m)$ as the highest power of 2 that divides m . If n is prime number greater than 3, then :

(A) $f(n^3 - 1) = f(n - 1)$

(B) $f(n^3 - 1) = f(n - 1) + 1$

(C) $f(n^3 - 1) = 2f(n - 1)$

(D) none of these is necessarily true

Ans. A

Sol. We can write $(n^3 - 1) = n^3 - 1^3$

Now, applying the formula $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$$\Rightarrow n^3 - 1 = (n - 1)(n^2 + n + 1)$$

We observe that $(n^3 - 1)$ is divisible by $(n - 1)$, so, if some 2^k divides $(n - 1)$, then that same 2^k will divide $(n^3 - 1)$ as $(n - 1)$ is a factor of $(n^3 - 1)$. We don't take into account $(n^2 + n + 1)$ as it is always odd, i.e., for both even and odd n , So, $(n^2 + n + 1)$ is not divisible by 2.

Therefore, $f(n^3 - 1) = f(n - 1)$.

This is true for all $n > 2$.

For $n = 2$,

$$(n^3 - 1) = (2^3 - 1) = 8 - 1 = 7$$

which is the only prime number of the form $(n^3 - 1)$. Since in question, they have clearly written that $n > 3$, the answer is $f(n^3 - 1) = f(n - 1)$

D02/1/018. Let $f(x) = \max. \{\sin t : 0 \leq t \leq x\}$

$$g(x) = \min. \{\sin t : 0 \leq t \leq x\}$$

$$\text{and } h(x) = [f(x) - g(x)]$$

where $[]$ denotes greatest integer function, then the range of $h(x)$ is :

(A) $\{0, 1\}$

(B) $\{1, 2\}$

(C) $\{0, 1, 2\}$

(D) $\{-3, -2, -1, 0, 1, 2, 3\}$

Ans. C

Sol.

D03/1/019. If $[\sin^{-1}(\cos^{-1}(\sin^{-1}(\tan^{-1}x)))] = 1$, where $[.]$ denotes greatest integer function, then complete set of values of x is :

(A) $[\tan(\sin(\cos 1)), \tan(\cos(\sin 1))]$

(B) $[\tan(\sin(\cos 1)), \tan(\sin(\cos(\sin 1)))]$

(C) $[\tan(\cos(\sin 1)), \tan(\sin(\cos(\sin 1)))]$

(D) $[\tan(\sin(\cos 1)), 1]$

Ans. B

Sol. $1 \leq \sin^{-1}(\cos^{-1}(\sin^{-1}(\tan^{-1}x))) \leq \frac{\pi}{2}$
 $\sin 1 \leq \cos^{-1}(\sin^{-1}(\tan^{-1}x)) \leq 1$
 $\cos(\sin 1) \geq \sin^{-1}(\tan^{-1}x) \geq \cos 1$
 $\sin(\cos(\sin 1)) \geq \tan^{-1}x \geq \sin(\cos 1)$
 $\tan(\sin(\cos(\sin 1))) \geq x \geq \tan(\sin(\cos 1))$

D03/1/020. Which of the following option(s) is/are equal to $\frac{\alpha}{\beta - \delta} + 4\gamma^2$, where

$$\alpha = 2 \tan^{-1} \sqrt{3 - 2\sqrt{2}} + \sin^{-1} \sqrt{\frac{1}{8 - 4\sqrt{3}}}, \beta = \cot^{-1}(\sqrt{3} - 2) + \frac{\sec^{-1}(-2)}{8},$$

$$\gamma = \sin \left(\frac{\pi}{\sqrt{3}} \tan \left(\tan^{-1}(\sqrt{2} - 1) + \cos^{-1} \frac{1}{2} \sqrt{2 + \sqrt{2 + \sqrt{3}}} \right) \right), \delta = \tan^{-1} \frac{1}{\sqrt{2}} + \cos^{-1} \frac{1}{\sqrt{3}} :$$

(A) 4 (B) 5 (C) 6 (D) 7

Ans. D

Sol.

D04/1/021. Let $a_n = \left(1 - \frac{1}{\sqrt{2}}\right) \dots \left(1 - \frac{1}{\sqrt{n+1}}\right), n \geq 1$. Then $\lim_{n \rightarrow \infty} a_n$:

(A) equals 1 (B) does not exist (C) equals $\frac{1}{\sqrt{\pi}}$ (D) equals 0

माना $a_n = \left(1 - \frac{1}{\sqrt{2}}\right) \dots \left(1 - \frac{1}{\sqrt{n+1}}\right), n \geq 1$ है। तो $\lim_{n \rightarrow \infty} a_n$:

(A) 1 के बराबर है (B) विद्यमान नहीं है (C) $\frac{1}{\sqrt{\pi}}$ के बराबर है (D) 0 के बराबर है

Ans. D

Sol. $a_n = \left(1 - \frac{1}{\sqrt{2}}\right) \dots \left(1 - \frac{1}{\sqrt{n+1}}\right)$

Now since,

$$\left(1 - \frac{1}{\sqrt{n+1}}\right) \leq \left(1 - \frac{1}{n+1}\right)$$

$$\Rightarrow \left(1 - \frac{1}{2}\right) \dots \left(1 - \frac{1}{n+1}\right) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \dots \frac{n}{n+1}$$

$$\Rightarrow a_n \leq \frac{1}{n+1}$$

Now, substitute

$$n \rightarrow \infty \Rightarrow a_n = 0$$

D04/1/022. The value of $\lim_{x \rightarrow 0} \frac{\ln(\sec(ex) \sec(e^2x) \dots \sec(e^{50}x))}{e^2 - e^{2\cos x}}$ is equal to :

$$\lim_{x \rightarrow 0} \frac{\ln(\sec(ex) \sec(e^2x) \dots \sec(e^{50}x))}{e^2 - e^{2\cos x}} \text{ का मान होगा :}$$

(A) $\frac{e^{100} - 1}{(e^2 - 1)}$

(B) $\frac{e^{100} - 1}{2(e^2 - 1)}$

(C) $\frac{2(e^{50} - 1)}{e^2 - 1}$

(D) $\frac{e^2(e^{100} - 1)}{2(e^2 - 1)}$

Ans. B

Sol. $\Rightarrow \frac{-[\ell n \cos ex + \ell n(\cos e^2x) + \dots + \ell n(\cos e^{50}x)]}{e^2 - e^{2\cos x}}$

Applying LHR

$$\Rightarrow \frac{\left[\frac{1}{\cos ex} \times \sin ex \cdot e + \dots + \frac{1}{\cos e^{50}x} \times \sin e^{50}x \times e^{50} \right]}{e^{2\cos x} \times 2 \sin x}$$

$$\Rightarrow \frac{[e \tan x + e^2 \tan e^2x + \dots + e^{50} \tan e^{50}x]}{2 \sin x e^{2\cos x}}$$

Applying LHR

$$\lim_{x \rightarrow 0} \frac{[e^2 \sec^2 ex + e^4 \sec^2 e^2x + \dots + e^{100} \sec^2 e^{50}x]}{-e^{2\cos x} \times 2 \sin^2 x + 2 \cos x e^{2\cos x}}$$

$$\Rightarrow \frac{[e^2 + e^4 + \dots + e^{100}]}{2e^2}$$

$$\Rightarrow \frac{[1 + e^2 + \dots + e^{50}]}{2}$$

$$\frac{(e^2)^{50} - 1}{2(e^2 - 1)} \Rightarrow \frac{e^{100} - 1}{2(e^2 - 1)}$$

D04/1/023. Which of the following options is/are correct, if :

$$S : \text{If } a_{n-1} = \frac{a_n - n}{n} \forall n \in \mathbb{N} - \{1\}, a_1 = 1, \text{ then } \lim_{n \rightarrow \infty} \prod_{r=1}^n \left(1 + \frac{1}{a_r}\right) = e.$$

$$T : \lim_{n \rightarrow \infty} \left(\frac{n^2 + n + 1}{n + 1} - (an + b) \right) = 4 \Rightarrow 7a - 2b = 15.$$

(A) S is True and T is False

(B) S is False and T is True

(C) Both S and T are True

(D) Both S and T are False.

निम्न में से कौनसा/से कथन सही है/हैं, यदि :

$$S : \text{यदि } a_{n-1} = \frac{a_n - n}{n} \forall n \in \mathbb{N} - \{1\}, a_1 = 1 \text{ हैं, तब } \lim_{n \rightarrow \infty} \prod_{r=1}^n \left(1 + \frac{1}{a_r}\right) = e.$$

$$T : \lim_{n \rightarrow \infty} \left(\frac{n^2 + n + 1}{n + 1} - (an + b) \right) = 4 \Rightarrow 7a - 2b = 15.$$

(A) S सत्य है तथा T गलत है

(B) S गलत तथा T सत्य है

(C) S तथा T दोनों सत्य हैं

(D) S तथा T दोनों गलत हैं

Ans. C

Sol.

D09/1/024. If $f''(x) + f'(x) + f^2(x) = x^2$ be the differential equation of a curve and let P be the point of maxima then number of tangents which can be drawn from P to $x^2 - y^2 = a^2$ is/are :

(A) 2

(B) 1

(C) 0

(D) either 1 or 2

Ans. A

Sol. At point of maxima $f'(x), f''(x) < 0$

$$f''(x) = x^2 - f^2(x) \leq 0$$

Points $P(x, f(x))$ lies outside $x^2 - y^2 = a^2$

Hence two tangents can be drawn.

D09/1/025. Let s be the set consisting of all those real numbers that can be written as $p - 2a$ where p and a are the

perimeter and area of a right-angled triangle having base length 1. The S is :

- (A) $(2, \infty)$ (B) $(1, \infty)$ (C) $(0, \infty)$ (D) The real line $\mathbb{R}$

Ans. A

Sol. Let x be the height of a triangle. Then $p = 1 + x + \sqrt{x^2 + 1}$, $a = \frac{x}{2}$. S is the set of all real numbers of the form

$$1 + x + \sqrt{x^2 + 1} - x = 1 + \sqrt{x^2 + 1} > 2.$$

D09/1/026. Let $a, b, c, d > 0$, be any real numbers. Then the maximum possible value of $cx + dy$, over all points on the

ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, must be :

- (A) $\sqrt{a^2c^2 + b^2d^2}$ (B) $\sqrt{a^2b^2 + c^2d^2}$ (C) $\sqrt{\frac{a^2c^2 + b^2d^2}{a^2 + b^2}}$ (D) $\sqrt{\frac{a^2b^2 + c^2d^2}{c^2 + d^2}}$

Ans. A

Sol. By the Cauchy-Schwarz inequality

$$cx + dy = ac \frac{x}{a} + bd \frac{y}{b} \leq \sqrt{a^2c^2 + b^2d^2} \sqrt{\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2} = \sqrt{a^2c^2 + b^2d^2}$$

Moreover, if

$$x = \frac{a^2c}{\sqrt{a^2c^2 + b^2d^2}}, y = \frac{b^2d}{\sqrt{a^2c^2 + b^2d^2}}$$

then

$$cx + dy = \sqrt{a^2c^2 + b^2d^2},$$

i.e. the maximum possible value is $\sqrt{a^2c^2 + b^2d^2}$

D09/1/027. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $\frac{d^2f(x)}{dx^2}$ is positive for all $x \in \mathbb{R}$, and suppose $f(0)$

$= 1$, $f(1) = 4$. Which of the following is not a possible value of $f(2)$:

- (A) 7 (B) 8 (C) 9 (D) 10

Ans. A

Sol. The derivative $f'(x)$ is strictly increasing. By the mid-point theorem, there exists $x_0 \in (0, 1)$, such that

$$f'(x_0) = \frac{f(1) - f(0)}{1} = 3.$$

So, for all $x \in [1, 2]$ we have $f'(x) > 3$ and $f(2) > f(1) + 3 = 7$.

7 is not a possible value of $f(2)$

D09/1/028. Let $g(x, y) = \max \{12 - x, 8 - y\}$. Then the minimum value of $g(x, y)$ as (x, y) varies over the line $x + y = 10$ is:

- (A) 5 (B) 7 (C) 1 (D) 3

Ans. A

Sol. Check if $12 - x$ is maximum.

$$\Rightarrow 12 - x > 8 - y$$

$$\text{but } y = 10 - x \because x + y = 10 \text{ (Given)}$$

$$\Rightarrow 12 - x > 8 - (10 - x)$$

$$\Rightarrow x < 7$$

Now,

$$g(x, y) = 12 - x, \text{ if } x < 7$$

Now, the minimum value of $12 - x$ is when x has maximum value that is 7

$$\Rightarrow \min(g(x, y)) = 12 - 7 = 5 \text{ if } x < 7$$

the minimum of $x - 2$ is when x has minimum value i.e. 7

$$\Rightarrow \min(g(x, y)) = 7 - 2 = 5 \text{ if } x > 7$$

$$\therefore \text{Answer} = 5$$

$\therefore$ A is the correct answer.

D06/1/029. Let $x = \frac{1+t}{t^3}$; $y = \frac{3}{2t^2} + \frac{2}{t}$ then the value of $\frac{dy}{dx} - x \left(\frac{dy}{dx} \right)^3$:

$$\text{माना } x = \frac{1+t}{t^3}; y = \frac{3}{2t^2} + \frac{2}{t} \text{ तब } \frac{dy}{dx} - x \left(\frac{dy}{dx} \right)^3 \text{ का मान होगा :}$$

- (A) 2 (B) 0 (C) -1 (D) -2

Ans. C

$$\text{Sol. } \frac{dx}{dt} = -\frac{3}{t^4} - \frac{2}{t^3} = \frac{-3-2t}{t^4}$$

$$\frac{dy}{dt} = \frac{-3}{t^3} - \frac{2}{t^2} = \frac{-3-2t}{t^3}$$

$$\frac{dy}{dx} = t$$

$$\frac{dy}{dx} - x \left(\frac{dy}{dx} \right)^3 = t - \left(\frac{1+t}{t^3} \right) \cdot t^3 = -1$$

D06/1/030. If $y^{-2} = 1 + 2\sqrt{2} \cos 2x$ then $\frac{d^2y}{dx^2} = y(py^2 + 1)(qy^2 - 1)$ then the value of $(p + q)$ equals to :

यदि $y^{-2} = 1 + 2\sqrt{2} \cos 2x$ तब $\frac{d^2y}{dx^2} = y(py^2 + 1)(qy^2 - 1)$ है तो $(p + q)$ का मान बराबर है :

- (A) 7 (B) 8 (C) 9 (D) 10

Ans. D

Sol. $-\frac{2}{y^3} y' = 2\sqrt{2} (-2 \sin 2x)$

$$\frac{(y')^2}{y^6} = 8 - (2\sqrt{2} \cos x)^2 = 8 - \left(\frac{1}{y^2} - 1\right)^2$$

$$\frac{(y')^2}{y^6} = \frac{8y^2 - (1 - y^2)^2}{y^4}$$

$(y')^2 = 8y^4 - y^2(1 - y^2)^2$ then diff.

D06/1/031. If $y = \tan \theta + \cot \theta$, $x = \tan \theta - \cot \theta$, where θ is a parameter then $y^3 \frac{d^2y}{dx^2}$ is equal to :

यदि $y = \tan \theta + \cot \theta$, $x = \tan \theta - \cot \theta$, जहाँ θ एक प्राचाल है तो $y^3 \frac{d^2y}{dx^2}$ बराबर है :

- (A) 2 (B) -2 (C) 4 (D) -4

Ans. C

Sol. By eliminating θ , we get

$$y^2 - x^2 = 4 \quad \Rightarrow 2y \frac{dy}{dx} - 2x = 0$$

$$\Rightarrow y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 1 \quad \Rightarrow y \frac{d^2y}{dx^2} + \frac{x^2}{y^2} = 1$$

$$y^3 \frac{d^2y}{dx^2} = y^2 - x^2 = 4$$

D06/1/032. If the normal to the curve $y = f(x)$ at $x = 0$ be given by $3x - y + 3 = 0$, then the value of

$$\lim_{x \rightarrow 1} (x-1)^2 \left\{ f\left((x-1)^2\right) - 5f\left(4(x-1)^2\right) + 4f\left(7(x-1)^2\right) \right\}^{-1} \text{ is :}$$

यदि $x = 0$ पर वक्र $y = f(x)$ का अभिलम्ब $3x - y + 3 = 0$ है तब

$\lim_{x \rightarrow 1} (x-1)^2 \left\{ f((x-1)^2) - 5f(4(x-1)^2) + 4f(7(x-1)^2) \right\}^{-1}$ का मान होगा :

- (A) $-\frac{1}{3}$ (B) $\frac{1}{3}$ (C) 3 (D) -3

Ans. A

Sol. Slope of normal = 3

$$\Rightarrow \text{slope of tangent } f'(0) = -\frac{1}{3}$$

$$\begin{aligned} L &= \lim_{x \rightarrow 1} \frac{(x-1)^2}{f((x-1)^2) - 5f(4(x-1)^2) + 4f(7(x-1)^2)} \quad \left(\frac{0}{0} \text{ form} \right) \\ &= \lim_{x \rightarrow 1} \frac{2(x-1)}{2(x-1)f'(x-1) - 40f'(4(x-1)^2)(x-1) + 56f'(7(x-1)^2)(x-1)} \\ &= \lim_{x \rightarrow 1} \frac{2}{2f'(x-1) - 40f'(4(x-1)^2) + 56f'(7(x-1)^2)} \\ &= \frac{1}{f'(0) - 20f'(0) + 28f'(0)} = \frac{1}{-\frac{1}{3} + \frac{20}{3} - \frac{28}{3}} = -\frac{1}{3} \end{aligned}$$

D06/1/033. $\lim_{x \rightarrow 1} (x-1) \ln(\sin^{-1}(\ln x))$ is equal to :

$\lim_{x \rightarrow 1} (x-1) \ln(\sin^{-1}(\ln x))$ बराबर है :

- (A) 0 (B) $\frac{\pi}{2}$ (C) $\ln \frac{\pi}{2}$ (D) 1

Ans. A

Sol. Let $x = 1 + t$

$$\lim_{t \rightarrow 0} \frac{\ln(\sin^{-1}(\ln(1+t)))}{\frac{1}{t}} \rightarrow \frac{\infty}{\infty}$$

$$\begin{aligned}
& \lim_{t \rightarrow 0} \frac{1}{\sin^{-1}(\ln(1+t))} \cdot \frac{1}{\sqrt{1-\ln^2(1+t)}} \cdot \frac{1}{(1+t)\left(-\frac{1}{t^2}\right)} \\
&= -\lim_{t \rightarrow 0} \frac{t^2}{\sin^{-1} \ln(1+t)} = -\lim_{t \rightarrow 0} \frac{t^2}{\sin^{-1} \ln(1+t)} \cdot \frac{\ln(1+t)}{\ln(1+t)} \\
&= -\lim_{t \rightarrow 0} t \cdot \frac{t}{\ln(1+t)} = 0
\end{aligned}$$

D06/1/034. $\lim_{x \rightarrow \infty} \frac{\cos^{-1}\left(\sqrt{x+\sqrt{x+\sqrt{x}}} - \sqrt{x+\sqrt{x}}\right)}{\sec^{-1}\left(-x^2 \ln\left(\frac{x^2+2}{x^2+1}\right)\right)}$ is equal to :

$$\lim_{x \rightarrow \infty} \frac{\cos^{-1}\left(\sqrt{x+\sqrt{x+\sqrt{x}}} - \sqrt{x+\sqrt{x}}\right)}{\sec^{-1}\left(-x^2 \ln\left(\frac{x^2+2}{x^2+1}\right)\right)} \text{ बराबर है :}$$

- (A) 1 (B) $\frac{\pi}{2}$ (C) $\frac{1}{2}$ (D) π

Ans. C

Sol. Numerator = $\lim_{x \rightarrow \infty} \cos^{-1}\left(\frac{\sqrt{x+\sqrt{x}} - \sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}} + \sqrt{x+\sqrt{x}}}\right)$

$$= \lim_{x \rightarrow \infty} \cos^{-1}\left(\frac{\sqrt{x}}{\left(\sqrt{x+\sqrt{x+\sqrt{x}}} + \sqrt{x+\sqrt{x}}\right)\left(\sqrt{x+\sqrt{x}} + \sqrt{x}\right)}\right)$$

$$= \cos^{-1}0 = \frac{\pi}{2}$$

$$\text{Denominator} = \lim_{x \rightarrow \infty} \sec^{-1} \left(-\ln \left(\frac{x^2 + 2}{x^2 + 1} \right)^{x^2} \right) = \sec^{-1} \left(-\lim_{x \rightarrow \infty} \frac{x^2}{x^2 + 1} \right) = \sec^{-1}(-1) = \pi$$

$$\therefore \text{Required limit} = \frac{\frac{\pi}{2}}{\pi} = \frac{1}{2}$$

D06/1/035. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that, $\lim_{\alpha \rightarrow \frac{\pi}{2}} g(2009 + \cos \alpha) = 3$ and

$$\lim_{n \rightarrow \infty} n \left(g \left(2009 + \frac{1}{n} \right) - g(2009) \right) = 27, \text{ then } \lim_{t \rightarrow 0^+} \left(\frac{g(2009)}{g(2009+t)} \right)^{\frac{1}{t}} \text{ is :}$$

माना $g : \mathbb{R} \rightarrow \mathbb{R}$ एक अवकलनीय फलन इस प्रकार है कि $\lim_{\alpha \rightarrow \frac{\pi}{2}} g(2009 + \cos \alpha) = 3$ तथा

$$\lim_{n \rightarrow \infty} n \left(g \left(2009 + \frac{1}{n} \right) - g(2009) \right) = 27, \text{ तब } \lim_{t \rightarrow 0^+} \left(\frac{g(2009)}{g(2009+t)} \right)^{\frac{1}{t}} \text{ है :}$$

(A) e^3

(B) e^2

(C) e^{-2}

(D) e^{-9}

Ans.

D

Sol.

$$\lim_{n \rightarrow \infty} \frac{g \left(2009 + \frac{1}{n} \right) - g(2009)}{\frac{1}{n}} = 27 \Rightarrow g'(2009)$$

$$= 27$$

$$\lim_{\alpha \rightarrow \frac{\pi}{2}} (2009 + \cos \alpha) = 3 \Rightarrow g(2009) = 3$$

$$\lim_{t \rightarrow 0^+} \left(\frac{g(2009)}{g(2009+t)} \right)^{\frac{1}{t}} \text{ is } 1^\infty \text{ form}$$

$$e^{\lim_{t \rightarrow 0^+} \frac{1}{t} \left(\frac{g(2009) - g(2009+t)}{g(2009+t)} \right)} \Rightarrow e^{-\lim_{t \rightarrow 0^+} \frac{1}{t} \left(\frac{g(2009+t) - g(2009)}{t} \right)} = e^{\frac{-g'(2009)}{g(2009)}} \Rightarrow e^{-9}$$

D06/1/036. Let $f: \mathbb{R} \rightarrow [0, \infty)$ be a continuous function such that $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. Suppose that f is

differentiable at $x=1$ and $\left. \frac{df(x)}{dx} \right|_{x=1} = 2$. Then, the value of $f(1) \log_e f(1)$ is :

माना $f: \mathbb{R} \rightarrow [0, \infty)$ में एक सतत् फलन इस प्रकार है कि $f(x+y) = f(x)f(y) \forall x, y \in \mathbb{R}$ है। माना कि f , $x=1$ पर

अवकलनीय है तथा $\left. \frac{df(x)}{dx} \right|_{x=1} = 2$ है। तब $f(1) \log_e f(1)$ का मान होगा :

- (A) e (B) 2 (C) $\log_e 2$ (D) 1

Ans. B

Sol. if $f(y)=0$ for some $y \in \mathbb{R}$, then for all $x \in \mathbb{R}$ we have $f(x) = f(x-y+y) = f(x-y)f(y) = 0$, which contradicts the assumption $f'(1) = 2$. So, $f(y) > 0$ for all $y \in \mathbb{R}$. Denote $g(x) = \log_e f(x)$. Then g is continuous and $g(x+y) = g(x) + g(y)$. It is well-known that $g(x) = cx$. So, $f(x) = e^{cx}$. By assumption, $f'(1) = ce^c = 2$. So, $f(1) \log_e f(1) = e^c c = 2$.

D06/1/037. Let $f(x-y) = \frac{f(x)}{f(y)}$ for all $x, y \in \mathbb{R}$ and $f'(0) = p, f'(5) = q$. Then the value of $f'(-5)$ is :

माना $f(x-y) = \frac{f(x)}{f(y)} \forall x, y \in \mathbb{R}$ तथा $f'(0) = p, f'(5) = q$ है। तब $f'(-5)$ का मान होगा :

- (A) q (B) $-q$ (C) $\frac{p}{q}$ (D) $\frac{p^2}{q}$

Ans. D

Sol. Observe that $f(0) = \frac{f(x)}{f(x)} = 1$. By assumption,

$$f(x) = f(x-y)f(y)$$

Differentiate this equation in x :

$$f'(x) = f'(x-y)f(y)$$

Differentiate in y

$$0 = -f'(x-y)f(y) + f(x-y)f'(y), f'(x-y)f(y) = f(x-y)f'(y)$$

Then

$$f(y) = \frac{f(0)f'(y)}{f'(0)} = \frac{f'(y)}{p}$$

Take $x=0, y=5$;

$$p = f'(0) = f'(-5)f(5) = \frac{f'(-5)f'(5)}{p} = \frac{q}{p}f'(-5), f'(-5) = \frac{p^2}{q}$$

D08/1/038. The number of distinct real roots of the polynomial $p(x) = (x^{2020} + 2020x^2 + 2020)(x^3 - 2020)(x^2 - 2020)$ is:

- (A) 2 (B) 3 (C) 2023 (D) 2025

Ans. B

Sol. Equation $x^{2020} + 2020x^2 + 2020 = 0$ has no real roots.

Equation $x^3 - 2020 = 0$ has one real root $x = \sqrt[3]{2020}$.

Equation $x^3 - 2020 = 0$ has two real roots $x = \pm\sqrt{2020}$.

D08/1/039. The polynomial $x^4 + 4x + c = 0$ has at least one real root if and only if:

- (A) $c < 2$ (B) $c \leq 2$ (C) $c < 3$ (D) $c \leq 3$

Ans. D

Sol. Consider equation $x^4 = a - 4x$ (i.e. $a = -c$). Consider the value of a when the line $y = a - 4x$ touches the curve $y = x^4$. If x_0 is the common point of these curves, then derivatives of functions $a - 4x$ and x^4 are equal at $x = x_0$:

$$-4 = 4x_0^3$$

Hence, $x_0 = -1$ and $(-1)^4 = a + 4$, $a = -3$. When $a \geq -3$ the curves $y = x^4$ and $y = a - 4x$ will intersect. In terms of c the answer is $c \leq 3$.

D08/1/040. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be any twice differentiable function such that its second derivative is continuous and $\frac{df(x)}{dx} \neq 0$

for all $x \neq 0$. If then $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \pi$:

- (A) for all $x \neq 0$, $f(x) > f(0)$ (B) for all $x \neq 0$, $f(x) < f(0)$
 (C) for all x , $\frac{d^2f(x)}{dx^2} > 0$ (D) for all x , $\frac{d^2f(x)}{dx^2} < 0$

Ans. A

Sol. Observe that $f(0) = 0$ (otherwise $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$ does not exist in $\mathbb{R}$). Since $\pi > 1$, there exists $\epsilon > 0$ such that

$$x \in (-\epsilon, \epsilon) \setminus \{0\} \Rightarrow f(x) > x^2$$

On the set $(0, \infty)$ we have $f'(x) \neq 0$. So, the derivative $f'(x)$ preserves the sign on $(0, \infty)$. If $x \in (0, \epsilon)$, then relations $f(0) = 0$, $f(x) > 0$ imply that f' is positive somewhere on $(0, x)$. But then it is positive everywhere

on $(0, \infty)$ and $f(x) > f(0)$ for $x > 0$.

On the set $(-\infty, 0)$ we have $f'(x) \neq 0$. So, the derivative $f'(x)$ preserves the sign on $(-\infty, 0)$. If $x \in (-\epsilon, 0)$, then relations $f(x) > 0$, $f(0) = 0$ imply that f' is negative somewhere on $(x, 0)$. But then it is negative everywhere on $(-\infty, 0)$ and $f(x) > f(0)$ for $x < 0$.

D08/1/041. If a, b are positive real variables whose sum is a constant λ , then the minimum value of $\sqrt{\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)}$ is

:

(A) $\frac{\lambda-1}{\lambda}$

(B) $\frac{\lambda+2}{\lambda}$

(C) $\frac{\lambda+1}{\lambda}$

(D) none of the above

Ans. B

Sol. Let $a + b = \lambda$

$$\therefore b = \lambda - a$$

$$\text{Let } f = \sqrt{\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)}$$

$$\text{Let } g = f^2 = \left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)$$

Taking derivative of g with respect to a and making it equal to 0, we get :

$$g' = \frac{-1}{a^2} \left(1 + \frac{1}{\lambda - a}\right) + \left(1 + \frac{1}{a}\right) \left(\frac{1}{(\lambda - a)^2}\right) = 0$$

Simplifying this we get :

$$(\lambda - 2a)(\lambda + 1) = 0, \text{ or } a = \frac{\lambda}{2}$$

Again on differentiating g with respect to a , we get :

$$g'' = \left(\frac{2}{a^3}\right) \left(1 + \frac{1}{\lambda - a}\right) - \frac{1}{a^2} \frac{1}{(\lambda - a)^2} + \left(1 + \frac{1}{a}\right) \left(\frac{2}{(\lambda - a)^3}\right) - \frac{1}{a^2} \left(\frac{1}{(\lambda - a)^2}\right)$$

$$\text{At } a = \frac{\lambda}{2} \text{ the value of } g'' > 0$$

$\Rightarrow$ It's a point of minima.

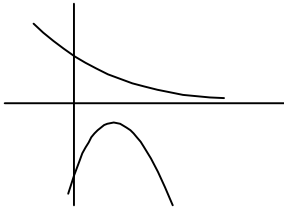
$$\therefore \text{Minimum value} = \frac{\lambda + 2}{\lambda}$$

D08/1/042. The number of real solutions of the equation $(9/10)^x = -3 + x - x^2$ is :

- (A) 2 (B) 0 (C) 1 (D) None of the above

Ans. B

Sol.



Above curves represent the graph of a^x where $0 < a < 1$ and the quadratic equation.

The quadratic equation is downward open parabola since the coefficient of x^2 is negative.

Vertex of the parabola is located at $x = \frac{-b}{2a} = \frac{1}{2}$

Also note that it will never intersect the x -axis as its Discriminant, D is -ve

$$D = b^2 - 4ac = 1^2 - 4(-1)(-3) = 1 - 12 = -11 < 0$$

Hence, the two curves will never intersect each other, which means no solution is possible to the given inequality.

$\therefore$ B is the correct option.

D08/1/043. $f(x)$ is a differentiable function on the real line such that $\lim_{x \rightarrow \infty} f(x) = 1$ and $\lim_{x \rightarrow \infty} f'(x) = \alpha$. Then :

- (A) α must be 0 (B) α need not be 0, but $|\alpha| < 1$
(C) $\alpha > 1$ (D) $\alpha < -1$

Ans. A

Sol.
$$f(n+1) - f(n) = \frac{f(n+1) - f(n)}{(n+1) - n} = f'(c_n), \quad c_n \in (n, n+1)$$

using the MVT (Lagrange's Theorem) for differentiable functions.

From this it follows that $c_n \xrightarrow{n \rightarrow \infty} \infty$, and thus

$$\alpha = \lim_{n \rightarrow \infty} f'(c_n) = \lim_{n \rightarrow \infty} (f(n+1) - f(n)) = 1 - 1 = 0$$

D08/1/044. Number of real solutions of the equation $x^7 + 2x^5 + 3x^3 + 4x = 2018$ is :

- (A) 1 (B) 3 (C) 5 (D) 7

Ans. A

Sol.

D08/1/045. If $t = \frac{\binom{200}{100}}{4^{100}}$, then :

- (A) $t < \frac{1}{3}$ (B) $\frac{1}{3} > t < \frac{1}{2}$ (C) $\frac{1}{2} < t < \frac{2}{3}$ (D) $\frac{2}{3} < t < 1$

Ans. A

Sol. Consider the function $f(n) = \frac{\binom{2n}{n}}{4^n}$

$$\text{So } f(n+1) = \frac{\binom{2n+2}{n+1}}{4^{n+1}}$$

Dividing the two and after solving, we get $\frac{f(n+1)}{f(n)} = \frac{\binom{n+1}{2}}{(n+1)} < 1 \forall n > 1$

That means $f(1) > f(2) > f(3) \dots > f(100)$

$$\Rightarrow f(100) < f(4) = \frac{70}{256} < \frac{1}{3}$$

Hence A is the correct option

D08/1/046. Let

$$A = \left\{ k \in \mathbb{R} \mid \alpha, \beta, \gamma \text{ are roots of } \frac{x^3}{a^2} + \left(a^2 + 4 + \frac{1}{a^2} \right) x = \frac{x^2}{a^2} + 1 \right\}$$

$$B = \left\{ \left| \frac{a}{2} - \frac{b}{2} \right| \mid |a-1| + |b-1| = |a| + |b| = |a+1| + |b+1|, ab \in \mathbb{R} \right\},$$

$$\lambda = \min(A), \mu = (\min(B)), k = \sum_{\alpha, \beta, \gamma} \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right).$$

Then which of the following options is/are equal to $\lambda + \mu$?

- (A) 5 (B) 2 (C) 3 (D) 4

Ans. D

Sol.

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

D05/2/047. Let $f(x) = x^2 \sin \frac{1}{x}$ for $0 < x \leq 1$ and $f(0) = 0$. If $g(x) = x^2$ for $x \in [0, 1]$ then which of the following statement(s) is/are correct ?

- (A) $f(x)$ is differentiable in $[0, 1]$ (B) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ does not exist
- (C) $\lim_{x \rightarrow 0} f'(x)g'(x)$ does not exist (D) $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$ does not exist

माना $f(x) = x^2 \sin \frac{1}{x} \forall 0 < x \leq 1$ तथा $f(0) = 0$ है। यदि $g(x) = x^2 \forall x \in [0, 1]$ है तब निम्न में से कौनसा/से कथन सही होगा/होंगे ?

- (A) $f(x)$, $[0, 1]$ में अवकलनीय है (B) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ विद्यमान नहीं है
- (C) $\lim_{x \rightarrow 0} f'(x)g'(x)$ विद्यमान नहीं है (D) $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$ विद्यमान नहीं है

Ans. ABD

Sol.

$$f'(0) = 0; f'(x) = \begin{cases} 2x \sin \frac{1}{x} - \cos \frac{1}{x}, & \text{for } 0 < x \leq 1 \\ 0, & \text{if } x = 0 \end{cases}$$

$\Rightarrow f(x)$ is differentiable in $[0, 1]$

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{x^2 \sin \left(\frac{1}{x} \right)}{x^2} = \text{D.N.E.}$$

$$\lim_{x \rightarrow 0} f'(x)g'(x) = \lim_{x \rightarrow 0} \left(4x^2 \sin \frac{1}{x} - 2x \cos \frac{1}{x} \right) = 0$$

Also, $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$ does not exist.

D05/2/048. Let f be a differentiable function satisfying $f(x+y) = f(x) + f(y) + (e^x - 1)(e^y - 1) \forall x, y \in \mathbb{R}$ and $f'(0) = 2$. Identify the correct statement(s) ?

$$(A) \lim_{x \rightarrow 0} \frac{f(f(x))}{f(x) - x} = 4$$

$$(B) \lim_{x \rightarrow 0} (f(x) + \cos x)^{\frac{1}{e^x - 1}} = e^2$$

(C) Number of solutions of the equation $f(x) = 0$ is 2

(D) Range of the function $y = f(x)$ is $(-\infty, \infty)$

माना f एक अवकलनीय फलन है जो $f(x + y) = f(x) + f(y) + (e^x - 1)(e^y - 1) \forall x, y \in \mathbb{R}$ को संतुष्ट करता है तथा $f'(0) = 2$ है। सही कथन/कथनों को चुनिए ?

$$(A) \lim_{x \rightarrow 0} \frac{f(f(x))}{f(x) - x} = 4$$

$$(B) \lim_{x \rightarrow 0} (f(x) + \cos x)^{\frac{1}{e^x - 1}} = e^2$$

(C) समीकरण $f(x) = 0$ के हलों की संख्या 2 है

(D) फलन $y = f(x)$ का परिसर $(-\infty, \infty)$ है

Ans. ABD

Sol. Partially differentiating w.r.t. x taking y as a constant, we get

$$f'(x + y) = f'(x) + e^x(e^y - 1)$$

$$\text{Put } x = 0, f'(y) = f'(0) + e^y - 1$$

$$f'(y) = e^y + 1$$

integrating both sides, we get

$$f(y) = e^y + y + c$$

$$\text{Putting } y = 0, 0 = 1 + 0 + c \Rightarrow c = -1$$

$$\therefore f(x) = e^x + x - 1$$

Now, verify the options.

D05/2/049. Let $f(x) = \sin^{-1}(2x\sqrt{1-x^2})$, then :

(A) $f(x)$ is continuous and differentiable at $x = \frac{1}{\sqrt{2}}$

(B) $f(x)$ is continuous and differentiable at $x = \frac{\pi}{4}$

(C) $f(x)$ is continuous but non-differentiable at $x = \frac{-1}{\sqrt{2}}$

(D) $f(x)$ is continuous but non-differentiable at $x = \frac{1}{\sqrt{2}}$

माना $f(x) = \sin^{-1}(2x\sqrt{1-x^2})$ है, तब :

(A) $f(x)$, $x = \frac{1}{\sqrt{2}}$ पर सतत् तथा अवकलनीय है

(B) $f(x)$, $x = \frac{\pi}{4}$ पर सतत् तथा अवकलनीय है

(C) $f(x)$, $x = \frac{-1}{\sqrt{2}}$ पर सतत् लेकिन अन-अवकलनीय

(D) $f(x)$, $x = \frac{1}{\sqrt{2}}$ पर सतत् लेकिन अन-अवकलनीय

Ans. BCD

Sol. Let $f(x) = \sin^{-1}(2x\sqrt{1-x^2})$

$$\text{Put } x = \sin \theta \Rightarrow \theta = \sin^{-1}x \in \left[\frac{-\pi}{2}, \frac{\pi}{2} \right]$$

$$\therefore y = \sin^{-1}(\sin 2\theta) = \begin{cases} -(\pi + 2\sin^{-1}x); & -1 \leq x \leq \frac{-1}{\sqrt{2}} \\ 2\sin^{-1}x; & \frac{-1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ (\pi - 2\sin^{-1}x); & \frac{1}{\sqrt{2}} < x \leq 1 \end{cases}$$

Clearly, $f(x)$ is continuous for all $x \in [-1, 1]$. But $f(x)$ is non-differentiable at $x = -1, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 1$

D05/2/050. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{[x]}{1+n^2} + \frac{3[x]}{2+n^2} + \frac{5[x]}{3+n^2} + \dots + \frac{(2n-1)[x]}{n+n^2} \right), & x \neq \frac{\pi}{2} \\ 1, & x = \frac{\pi}{2} \end{cases}$ where

$[y]$ denotes largest integer $\leq y$, then which of the following statement(s) is(are) correct ?

(A) $f(x)$ is injective but not surjective

(B) $f(x)$ is non-derivable at $x = \frac{\pi}{2}$

(C) $f(x)$ is discontinuous at all integers and continuous at $x = \frac{\pi}{2}$

(D) $f(x)$ is unbounded function

माना $f : \mathbb{R} \rightarrow \mathbb{R}$ इस प्रकार परिभाषित है कि

$$f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{[x]}{1+n^2} + \frac{3[x]}{2+n^2} + \frac{5[x]}{3+n^2} + \dots + \frac{(2n-1)[x]}{n+n^2} \right), & x \neq \frac{\pi}{2} \\ 1, & x = \frac{\pi}{2} \end{cases} \quad \text{जहाँ } [y] \text{ महत्तम पूर्णांक } \leq y \text{ को दर्शाता है}$$

है, तो निम्न में से कौनसा/से कथन सही होगा/होंगे ?

(A) $f(x)$ एकैकी है लेकिन आच्छादक नहीं

(B) $f(x)$, $x = \frac{\pi}{2}$ पर अन-अवकलनीय है

(C) $f(x)$ सभी पूर्णाकों के लिए असतत् है तथा $x = \frac{\pi}{2}$ पर सतत् है

(D) $f(x)$ अपरिमित फलन है

Ans. CD

Sol.

$$f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{1}{1+n^2} + \frac{3}{2+n^2} + \dots + \frac{(2n-1)}{n+n^2} \right) [x], & x \neq \frac{\pi}{2} \\ 1, & x = \frac{\pi}{2} \end{cases}$$

$$\therefore \frac{1}{n+n^2} + \frac{3}{n+n^2} + \dots + \frac{(2n-1)}{(n+n^2)} < \frac{1}{1+n^2} + \frac{3}{2+n^2} + \dots + \frac{(2n-1)}{(n+n^2)} <$$

$$\frac{1}{1+n^2} + \frac{3}{1+n^2} + \dots + \frac{(2n-1)}{1+n^2}$$

$$\Rightarrow \frac{n^2}{n+n^2} < \frac{1}{1+n^2} + \frac{3}{2+n^2} + \dots + \frac{(2n-1)}{n+n^3} < \frac{n^2}{1+n^2}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{n^2}{n+n^2} = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{n^2}{1+n^2} = 1$$

$$\therefore \lim_{n \rightarrow \infty} \left(\frac{1}{1+n^2} + \frac{3}{2+n^2} + \dots + \frac{(2n-1)}{n+n^2} \right) = 1$$

$$\therefore f(x) = \begin{cases} [x], & x \neq \frac{\pi}{2} \\ 1, & x = \frac{\pi}{2} \end{cases}$$

$\Rightarrow f(x) = [x] \forall x \in \mathbb{R}$. Hence range of $f(x)$ is set of all integers.

$\Rightarrow f(x)$ is unbounded.

Now, at $x = \frac{\pi}{2}$,

Clearly L.H.L. = R.H.L. = $1 = f\left(\frac{\pi}{2}\right)$

But $f(x)$ is discontinuous at all integers as $[x]$ is discontinuous at integers.

D05/2/051. If $f(x) = \tan^{-1}(\operatorname{sgn}(x^2 - \lambda x + 1))$ has exactly one point of discontinuity, then the value of λ can be :

यदि $f(x) = \tan^{-1}(\operatorname{sgn}(x^2 - \lambda x + 1))$ का केवल एक असततता बिन्दु है, तब λ का मान होगा :

- (A) 1 (B) -1 (C) 2 (D) -2

Ans. CD

Sol. $f(x)$ has exactly one point of discontinuity so that $\operatorname{sgn}(x^2 - \lambda x + 1)$ is equal to zero for some values of λ .

$\therefore D = 0$

$\Rightarrow \lambda^2 - 4 = 0$

$\Rightarrow \lambda = \pm 2$

D05/2/052. The function $f(x) = \left[x^2 \left[\frac{1}{x^2} \right] \right]$, $x \neq 0$ is ($[x]$ represents the greatest integer $\leq x$) :

- (A) Continuous at $x = 1$ (B) Discontinuous at $x = -1$
(C) Discontinuous at infinitely many points (D) Continuous everywhere

फलन $f(x) = \left[x^2 \left[\frac{1}{x^2} \right] \right]$, $x \neq 0$ है ($[x]$ महत्तम पूर्णांक $\leq x$ को दर्शाता है) :

- (A) $x = 1$ पर सतत् (B) $x = -1$ पर असतत्
(C) अनन्त बिन्दुओं पर असतत् (D) सभी जगह सतत्

Ans. BC

Sol. We have, $f(1) = f(-1) = 1$

Let $x > 1$, then $0 < \frac{1}{x^2} < 1 \Rightarrow \left[\frac{1}{x^2} \right] = 0 \Rightarrow f(x) = 0 \forall x > 1$

Also, if $x < -1$, then $x^2 > 1 \Rightarrow 0 < \frac{1}{x^2} < 1 \Rightarrow f(x) = 0 \forall x < -1$

Hence, $f(1) = 1$, $f(-1) = 1$ and $f(x) = 0$ if $|x| > 1$

$\therefore f(x)$ cannot be continuous at $x = 1$ and $x = -1$

Again,

$$\text{Let } \frac{1}{2} < x^2 < 1 \Rightarrow 1 < \frac{1}{x^2} < 2 \Rightarrow \left\lfloor \frac{1}{x^2} \right\rfloor = 1 \Rightarrow \frac{1}{2} < x^2 \left\lfloor \frac{1}{x^2} \right\rfloor < 1$$

$$\therefore \left\lfloor x^2 \left\lfloor \frac{1}{x^2} \right\rfloor \right\rfloor = 0 \Rightarrow f(x) = 0 \text{ if } x \in \left(-1, \frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$$

$$\text{Next, let } \frac{1}{3} < x^2 < \frac{1}{2} \Rightarrow 2 < \frac{1}{x^2} < 3 \Rightarrow \left\lfloor \frac{1}{x^2} \right\rfloor = 2 \Rightarrow \frac{2}{3} < x^2 \left\lfloor \frac{1}{x^2} \right\rfloor < 1$$

$$\therefore \left\lfloor x^2 \left\lfloor \frac{1}{x^2} \right\rfloor \right\rfloor = 0 \Rightarrow f(x) = 0, \text{ if } x \in \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{3}}\right) \cup \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{2}}\right)$$

$$\text{At } x = \pm \frac{1}{\sqrt{2}}, x^2 = \frac{1}{2}, \frac{1}{x^2} = 2 \Rightarrow f(x) = 1$$

$$\text{Similarly at } x = \frac{1}{\sqrt{3}}, \frac{1}{2}, \frac{1}{\sqrt{5}}, \dots$$

$$\therefore f(x) \text{ is discontinuous at infinite number of points given by } x \in \left\{ \pm \frac{1}{\sqrt{n}}, n \in \mathbb{N} \right\}.$$

D05/2/053. Which of the following statements is/are correct ?

[Note : $[x]$ and $\{x\}$ denote greatest integer less than or equal to x and fractional part of x respectively] :

(A) $f(x) = [\ln x] + \sqrt{\{\ln x\}}, x > 1$ is continuous at $x = e$

(B) If $\lim_{x \rightarrow -2} \left(\frac{3x^2 + ax + a + 1}{x^2 + x - 2} \right)$ exists and equals l , then $a + \frac{1}{l} = 10$

(C) $f: [-1, 1] \rightarrow [-1, 1]$, $f(x) = x^2 \operatorname{sgn}(x)$ is a bijective function, where $\operatorname{sgn} x$ denotes signum function of x .

(D) If f is continuous on $[-1, 1]$, $f(-1) = 4$ and $f(1) = 3$, then there exists a number r such that $|r| < 1$ and $f(r) = \pi$

निम्न में से कौनसे कथन सत्य है :

[Note : $[x]$ तथा $\{x\}$ क्रमशः महत्तम पूर्णांक फलन तथा आंशिक भाग फलन को निरूपित करते हैं] :

(A) $f(x) = [\ln x] + \sqrt{\{\ln x\}}, x > 1$, $x = e$ पर सतत है

(B) यदि $\lim_{x \rightarrow 2} \left(\frac{3x^2 + ax + a + 1}{x^2 + x - 2} \right)$ विद्यमान है तथा l के बराबर है, तो $a + \frac{1}{l} = 10$

(C) $f: [-1, 1] \rightarrow [-1, 1]$, $f(x) = x^2 \operatorname{sgn} x$ एक एकैकी फलन है, जहाँ $\operatorname{sgn} x$, x के सिग्नम फलन को निरूपित करता है

(D) यदि $f, [-1, 1]$ में सतत है, $f(-1) = 4$ तथा $f(1) = 3$ है, तो संख्या r इस प्रकार विद्यमान है, कि $|r| < 1$ तथा $f(r) = \pi$

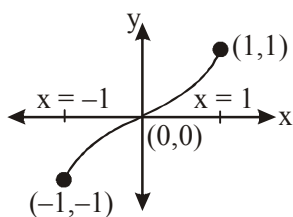
Ans.

ABCD

Sol.

(A) $f(e) = 1$; $f(e^+) = 1$; $f(e^-) = 1$

(B) Here, $a = 13$, $l = \frac{-1}{3}$; so $a + \frac{1}{l} = 13 + \left(\frac{1}{\frac{-1}{3}} \right) = 13 - 3 = 10$



$$(C) f(x) = x^2 \operatorname{sgn} x = \begin{cases} -x^2, & x < 0 \\ 0, & x = 0 \\ x^2, & x > 0 \end{cases}$$

(D) As, f is continuous on $[-1, 1]$ and $f(-1) = 4$, $f(1) = 3$

$\Rightarrow$ there exists a number r such that $|r| < 1$ and $f(r) = \pi$.

D05/2/054. If $f(x)$ be such that $f(x) = \max \{ |2 - x|, 2 - x^3 \}$, then :

(A) $f(x)$ is continuous $\forall x \in \mathbb{R}$

(B) $f(x)$ is differentiable $\forall x \in \mathbb{R}$

(C) $f(x)$ is non-differentiable at one point only

(D) $f(x)$ is non-differentiable at 4 points only

यदि $f(x)$ इस प्रकार है, कि $f(x) = \max \{ |2 - x|, 2 - x^3 \}$, तो :

(A) x के सभी वास्तविक मानों के लिये $f(x)$ सतत होगा

(B) x के सभी वास्तविक मानों के लिए $f(x)$ अवकलनीय होगा

(C) $f(x)$ केवल एक बिन्दु पर अवकलनीय नहीं होगा

(D) $f(x)$ केवल 4 बिन्दुओं पर अवकलनीय नहीं होगा

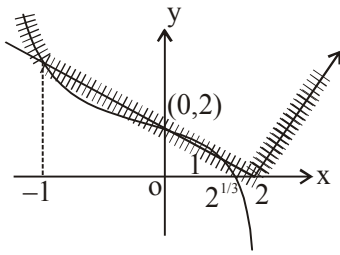
Ans.

AD

Sol.

$f(x) = \max \{ |2 - x|, 2 - x^3 \}$, then

$f(x)$ is continuous $\forall x \in \mathbb{R}$



and $f(x)$ is non-differentiable at $x = \pm 1, 0, 2$

$$f(x) = \max \{ |2 - x|, 2 - x^3 \},$$

D05/2/055. Consider $f(x) = \begin{cases} -x^3 & , -1 \leq x \leq 0 \\ x & , 0 < x < 1 \\ 2-x & , 1 \leq x \leq 2 \\ (x-2)^2 & , x > 2 \end{cases}$, $g(x) = \max \{ f(t), -1 \leq t \leq x \}, -1 \leq x < \infty$

Which of the following holds good ?

- (A) $g(x)$ is continuous function $\forall x \in [-1, \infty)$ (B) $g(x)$ is derivable function $\forall x \in \mathbb{R}$
 (C) range of $g(x)$ is $[1, \infty)$ (D) $g(x)$ is many one function

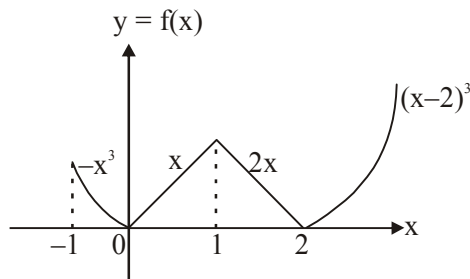
माना $f(x) = \begin{cases} -x^3 & , -1 \leq x \leq 0 \\ x & , 0 < x < 1 \\ 2-x & , 1 \leq x \leq 2 \\ (x-2)^2 & , x > 2 \end{cases}$, $g(x) = \text{अधिकतम } \{ f(t), -1 \leq t \leq x \}, -1 \leq x < \infty$ है।

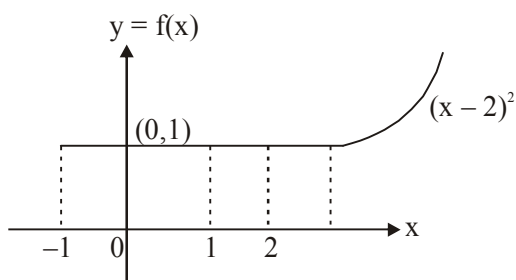
निम्न में से कौनसा सत्य होगा ?

- (A) $g(x)$, $\forall x \in [-1, \infty)$ संतत फलन होगा (B) $g(x)$, $\forall x \in \mathbb{R}$ अवकलनीय फलन होगा
 (C) $g(x)$ का परिसर $[1, \infty)$ होगा (D) $g(x)$ बहुएकैकी फलन होगा

Ans. ACD

Sol. 3.





D05/2/056. Let $f(x) = \begin{cases} \left(\cos \frac{1}{x}\right) \ln^2(1+x) & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$, then which of the following is(are) correct :

- (A) $f'(x)$ is non differentiable at $x = 0$ (B) $f(x)$ is differentiable at $x = 0$
 (C) $f'(x)$ is continuous at $x = 0$ (D) $f'(x)$ is discontinuous at $x = 0$

माना $f(x) = \begin{cases} \left(\cos \frac{1}{x}\right) \ln^2(1+x) & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$, तब निम्न में से कौनरसा/से सही होगा/होंगे :

- (A) $f'(x)$, $x = 0$ पर अवकलनीय नहीं है (B) $f(x)$, $x = 0$ पर अवकलनीय है
 (C) $f'(x)$, $x = 0$ पर सतत् है (D) $f'(x)$, $x = 0$ पर असतत् है

Ans. ABD

Sol. $f(x) = \begin{cases} \left(\cos \frac{1}{x}\right) \cdot \ln^2(1+x) & x > 0 \\ 0 & x \leq 0 \end{cases}$
 $f(0+h) = f(0) = f(0-h) = 0 \Rightarrow f(x)$ is continuous

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cos\left(\frac{1}{h}\right) \cdot \ln^2(1+h)}{h} = 0$$

$\Rightarrow f(x)$ is differentiable at $x = 0$

$$f'(x) = \sin\left(\frac{1}{x}\right) \cdot \frac{1}{x^2} \ln^2(1+x) + \cos\left(\frac{1}{x}\right) \cdot \frac{2 \ln(1+x)}{1+x}, x > 0$$

$$= 0, \quad x \leq 0$$

$$f'(0^+) = 0 = f'(0^-) = f'(0)$$

$\Rightarrow f'(x)$ is continuous at $x = 0$

D05/2/057. If $f(x) = \lim_{t \rightarrow 0} \frac{|a + \sin \pi x|^t - 1}{|a + \sin \pi x|^t + 1}$, $x \in (0, 6)$, then :

- (A) If $a = 1$, then $f(x)$ has 5 points of discontinuity
- (B) If $a = 3$, then $f(x)$ has not point of discontinuity
- (C) If $a = 0.5$, then $f(x)$ has 6 points of discontinuity
- (D) If $a = 0$, then $f(x)$ has 6 points of discontinuity

यदि $f(x) = \lim_{t \rightarrow 0} \frac{|a + \sin \pi x|^t - 1}{|a + \sin \pi x|^t + 1}$, $x \in (0, 6)$, तब :

- (A) यदि $a = 1$, तब $f(x)$ के 5 असततता बिन्दु हैं
- (B) यदि $a = 3$, तब $f(x)$ के कोई असततता बिन्दु नहीं हैं
- (C) यदि $a = 0.5$, तब $f(x)$ के 6 असततता बिन्दु हैं
- (D) यदि $a = 0$, तब $f(x)$ के 6 असततता बिन्दु हैं

Ans. ABCD

Sol. Given function is discontinuous when $a + \sin \pi x = \pm 1$

Now, if $a = 1$, then $\sin \pi x = 0$ or $x = 1, 2, 3, 4, 5$.

If $a = 3$, then $\sin \pi x = -2$, not possible.

If $a = 0.5$, then $\sin \pi x = 0.5$

Therefore, x has 6 value, 2 each for one cycle of period 2.

If $a = 0$, then $\sin \pi x = \pm 1$ or $x = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}, \frac{11}{2}$

D05/2/058. If number of points of discontinuity of the function $f(x) = [2 + 10 \sin x]$, in $x \in \left[0, \frac{\pi}{2}\right]$ is same as number of points of non-differentiability of the function

$g(x) = (x - 1) \times (x - 2) |(x - 1)(x - 2) \dots (x - 2m)|$, ($m \in \mathbb{N}$) in $x \in (-\infty, \infty)$, then the value of m is :

यदि फलन $f(x) = [2 + 10 \sin x]$ के $x \in \left[0, \frac{\pi}{2}\right]$ में असततता बिन्दुओं की संख्या फलन $g(x) = (x - 1) \times (x - 2) |(x - 1)(x - 2) \dots (x - 2m)|$, ($m \in \mathbb{N}$) के $x \in (-\infty, \infty)$ में अनवकलनीयता बिन्दुओं की संख्या के समान है, तब m का मान होगा :

- (A) 4
- (B) 5
- (C) 6
- (D) 7

Ans. C

Sol. $2 + 10 \sin x \in [2, 12]$

Clearly, $f(x)$ is discontinuous where $2 + 10 \sin x = 3, 4, 5, \dots, 12$, i.e., at 10 points.

Number of points of non-differentiability of $g(x) = (x-1)(x-2)\dots(x-2m)$ is $(2m-2)$ as $g(x)$ is differentiable at $x=1$ and $x=2$.

$$\therefore 2m-2=10$$

$$\therefore m=6$$

D05/2/059. Let $[x] = \max \{a \in \mathbb{Z} : a \leq x\}$, Then which of the following options is/are correct ?

- (A) If f is continuous function such that $[f(-1)] = 2$, $[f(0)] = -2$, then $f(x) = 0$ has at least one solution in $(-1, 0)$.
- (B) If f is discontinuous at $x = a$, and g is continuous at $x = a$, $g(a) = 0$, then $f(x)g(x)$ must be continuous at $x = a$.
- (C) If $f(x) + \sin x$ is continuous and $g(x) - e^x$ is discontinuous at $x = a$, then $f(x)g(x)$ must be discontinuous at $x = a$.
- (D) It is possible that $f(x)$ and $g(x) = \max \{t \in \mathbb{Z} : t \leq f(x)\}$ are both continuous at $x = a$, $f(a) \in \mathbb{Z}$.

माना $[x] = \max \{a \in \mathbb{Z} : a \leq x\}$ है। तब कौनसा/से विकल्प सही होगा/होंगे ?

- (A) यदि f एक सतत फलन इस प्रकार है कि $[f(-1)] = 2$, $[f(0)] = -2$, तब $f(x) = 0$ का $(-1, 0)$ में कम से कम एक हल होगा।
- (B) यदि f , $x = a$ पर एक असतत फलन है तथा g , $x = a$ पर सतत है, $g(a) = 0$ है, तब $f(x)g(x)$, $x = a$ पर सतत होगा।
- (C) यदि $x = a$ पर $f(x) + \sin x$ सतत तथा $g(x) - e^x$ असतत है, तो $f(x)g(x)$, $x = a$ पर असतत होगा।
- (D) यह संभव है कि $f(x)$ तथा $g(x) = \max \{t \in \mathbb{Z} : t \leq f(x)\}$, $x = a$, $f(a) \in \mathbb{Z}$ पर दोनों सतत हैं

Ans. AD

Sol.

D05/2/060. Which of the following options is/are correct, if

$$f(x) = \lim_{n \rightarrow \infty} \frac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}; n \in \mathbb{N}, \text{ and}$$

$$A = \{a \in \mathbb{Z} : g(x) = [x]^2 - [x^2] \text{ is continuous at } x = a.\}$$

Here $|X|$ denotes the cardinal number of set X :

- (A) $\lim_{x \rightarrow 1^+} f(x) = -1$ (B) $\lim_{x \rightarrow 1^-} f(x) = -1$
- (C) f is discontinuous at $x = 1$ (D) $|A| = 0$

निम्न में से कौनसा/से विकल्प सही होगा/होंगे, यदि

$$f(x) = \lim_{n \rightarrow \infty} \frac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}; n \in \mathbb{N}, \text{ तथा}$$

$$A = \{a \in \mathbb{Z} : g(x) = \lfloor x \rfloor^2 - \lfloor x^2 \rfloor, x = a \text{ पर सतत है}\}$$

यहाँ $|X|$ समुच्चय X की कार्डिनल संख्या को दर्शाता है :

$$(A) \lim_{x \rightarrow 1^+} f(x) = -1$$

$$(B) \lim_{x \rightarrow 1^-} f(x) = -1$$

$$(C) f, x = 1 \text{ पर असतत है}$$

$$(D) |A| = 0$$

Ans. AC

Sol.

D05/2/061. Let $[x] = \max \{a \in \mathbb{Z} : a \leq x\}, \{x\} = x - [x],$

$$f(x) = [x]^{\{x\}} + \{x\}^{[x]}; f'\left(\frac{5}{2}\right) = 1 + \sqrt{\lambda} \ln 2, g(x) = \begin{cases} b \sin^{-1} \frac{x+c}{2}, & -\frac{1}{2} < x < 0 \\ \frac{1}{2}, & x = 0 \\ \frac{e^{\frac{ax}{2}} - 1}{x}, & 0 < x < \frac{1}{2} \end{cases}.$$

If g is differentiable at $x = 0$ and $|c| < \frac{1}{2}$, then :

$$\text{माना } [x] = \max \{a \in \mathbb{Z} : a \leq x\}, \{x\} = x - [x]$$

$$f(x) = [x]^{\{x\}} + \{x\}^{[x]}; f'\left(\frac{5}{2}\right) = 1 + \sqrt{\lambda} \ln 2, g(x) = \begin{cases} b \sin^{-1} \frac{x+c}{2}, & -\frac{1}{2} < x < 0 \\ \frac{1}{2}, & x = 0 \\ \frac{e^{\frac{ax}{2}} - 1}{x}, & 0 < x < \frac{1}{2} \end{cases}.$$

यदि $g, x = 0$ पर अवकलनीय है तथा $|c| < \frac{1}{2}$, तब :

$$(A) a(4 - c^2) = \lambda^6 b^2 \quad (B) a(4 - c^2) = \lambda^{12} b^2 \quad (C) a(\lambda^2 - c^2) = \lambda^6 b^2 \quad (D) a(4 - c^2) = \lambda^3 b^2$$

Ans. AC

Sol.

D02/2/062. Let $P(x) = x^2 + bx + c$ where $b, c \in \mathbb{R}$. If $P(P(1)) = 0 = P(P(2))$ and $P(1) \neq P(2)$. Then :

माना $P(x) = x^2 + bx + c$, जहाँ $b, c \in \mathbb{R}$ है। यदि $P(P(1)) = 0 = P(P(2))$ तथा $P(1) \neq P(2)$ है, तो

(A) $2b + 1 = 0$ (B) $2c + 3 = 0$ (C) $2b + 3 = 0$ (D) $2c + 1 = 0$

Ans. AB

Sol. Since $P(P(1)) = 0$ and $P(P(2)) = 0$

$\therefore P(1)$ and $P(2)$ are the roots of $P(x) = 0$

$$P(x) = x^2 - x(P(1) + P(2)) + P(1)P(2)$$

$$\therefore b = -(P(1) + P(2)) = -(5 + 3b + 2c) \dots (1)$$

$$c = P(1)P(2) = (1 + b + c)(4 + 2b + c) \dots (2)$$

from (1) and (2), we get $b = \frac{-1}{2}$ and $c = \frac{-3}{2}$.

D02/2/063. Which of the following is/are CORRECT ?

(A) $y = \ln(x + \sqrt{1+x^2})$ is an odd function

(B) $y = \frac{x}{2[x]+1} (x \neq 1)$ is an even function ([.] is GIF)

(C) $y = \frac{2e^x}{1+e^x} - 2x - 1$ is an odd function

(D) If $y = f(x)$ is odd and $y = g(x)$ is even then $y = f(g(x))$ is an even function

निम्न में से कौनसा/से सत्य है/हैं ?

(A) $y = \ln(x + \sqrt{1+x^2})$ एक विषम फलन है

(B) $y = \frac{x}{2[x]+1} (x \neq 1)$ एक सम फलन है ([.] महत्तम पूर्णांक फलन है)

(C) $y = \frac{2e^x}{1+e^x} - 2x - 1$ एक विषम फलन है

(D) यदि $y = f(x)$ विषम तथा $y = g(x)$ सम है तब $y = f(g(x))$ एक सम फलन है

Ans. ABCD

Sol. (A) $f(x) + f(-x) = 0$, odd

$$(B) f(-x) = \frac{-x}{2(-1-[x])+1} = \frac{-x}{-2-2[x]+1} = \frac{x}{1+2[x]} = f(x), \text{ even}$$

$$(C) \quad f(-x) = \frac{2e^{-x}}{1+e^{-x}} + 2x - 1$$

$$f(x) + f(-x) = \frac{2}{1+e^x} + 2x - 1 + \frac{2e^x}{1+e^x} - 2x - 1 = 0, \text{ odd}$$

$$(D) \quad y = f(x), \text{ odd}$$

$$y = g(x), \text{ even}$$

$$y = f(g(-x)) = f(g(x)) = \text{even}$$

D02/2/064. Which of the following statement(s) is/are correct ?

(A) Minimum value of $f(x) = (x^2 - 5x + 9)(x^2 - 5x + 3)$ is -9

(B) Graph of function $f(x) = \operatorname{sgn}\left(\cos\left(\frac{1}{e^x + e^{-x}}\right)\right)$ is symmetric with respect to origin, where $\operatorname{sgn}(x)$ denotes signum function of x .

(C) The domain of function $f(x) = \sqrt{1 - \sqrt{2 - \sqrt{3 - x}}}$ is $[-1, 2]$

(D) $f: (-\infty, 0] \rightarrow [0, \infty)$ defined by $f(x) = \frac{e^{-[x]}}{e^{\{x\}}} - e^{|x|}$ is one-one function

(where $[.]$ is greatest integer function and $\{.\}$ is fractional part function)

निम्न में से कौन/कौनसे कथन सही है –

(A*) $f(x) = (x^2 - 5x + 9)(x^2 - 5x + 3)$ का न्यूनतम मान -9 है।

(B) फलन $f(x) = \operatorname{sgn}\left(\cos\left(\frac{1}{e^x + e^{-x}}\right)\right)$ का आरेख मूल बिन्दु के सापेक्ष सममित है, जहाँ $\operatorname{sgn}(x)$ सिग्नेम फलन को निरूपित करती है।

(C*) फलन $f(x) = \sqrt{1 - \sqrt{2 - \sqrt{3 - x}}}$ का प्रांत $[-1, 2]$ है।

(D) $f: (-\infty, 0] \rightarrow [0, \infty)$ जहाँ $f(x) = \frac{e^{-[x]}}{e^{\{x\}}} - e^{|x|}$ एकैकी फलन है।

(जहाँ $[.]$ महतम पूर्णांक फलन तथा $\{.\}$ भिन्नात्मक भाग (fractional part) फलन है)

Ans. AC

Sol. (A) $f(x) = (x^2 - 5x + 9)(x^2 - 5x + 3)$

$$\text{Let } x^2 - 5x + 9 = t$$

$$f(t) = t(t - 6) = t^2 - 6t + 9 - 9$$

$$f_{\min} = -9$$

$$(B) f(x) = \operatorname{sgn}\left(\cos\left(\frac{1}{e^x + e^{-x}}\right)\right) = 1$$

$$(C) f(x) = \sqrt{1 - \sqrt{2 - \sqrt{3 - x}}}$$

$$1 - \sqrt{2 - \sqrt{3 - x}} \geq 0$$

$$1 \geq 2 - \sqrt{3 - x}$$

$$\sqrt{3 - x} \geq 1$$

$$3 - x \geq 1$$

$$x \leq 2 \quad \dots(i)$$

$$2 - \sqrt{3 - x} \geq 0$$

$$2 \geq \sqrt{3 - x}$$

$$4 \geq 3 - x$$

$$x \geq -1 \quad \dots(ii)$$

$$x \in [-1, 2]$$

$$(D) f(x) = e^{-x} - e^{|x|} = \begin{cases} e^{-x} - e^x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Many one

D02/2/065. Let $f(x) = x^2 - 2bx + b(b+1)$; $f: [b, \infty) \rightarrow [b, \infty)$. If one of the solution of the equation $f(x) = f^{-1}(x)$ is 2009; then other solution(s) is/are :

माना $f(x) = x^2 - 2bx + b(b+1)$; $f: [b, \infty) \rightarrow [b, \infty)$ है। यदि समीकरण $f(x) = f^{-1}(x)$ का एक हल 2009 है; तब दूसरा हल होगा/होंगे :

(A) 2008 (B) 2010 (C) 0 (D) 1

Ans. AB

Sol. $f: [b, \infty) \rightarrow [b, \infty)$ and $f(x) = (x-b)^2 + b$

$$\Rightarrow f^{-1}(x) = b + \sqrt{x-b}$$

$$\therefore f(x) = f^{-1}(x) \Rightarrow (x-b)^2 = \sqrt{x-b} \Rightarrow x-b=0 \text{ or } x-b=1$$

$$\Rightarrow x=b \text{ Or } x=b+1$$

$$\Rightarrow \text{difference of both values of } x \text{ is } 1.$$

D02/2/066. If P and Q are the points of intersection of $y = g(x)$ and $y = g^{-1}(x)$ then which of the following statement(s)

is/are incorrect?

- (A) P and Q necessarily lie on the line $y = x$
 (B) P and Q must be coincident
 (C) Slope of line joining P and Q is necessarily zero
 (D) Slope of line joining P and Q may be -1

यदि P तथा Q वक्रों $y = g(x)$ तथा $y = g^{-1}(x)$ के प्रतिछेदन बिन्दु है, तो निम्न में से कौनसे कथन असत्य है—

- (A) P तथा Q रेखा $y = x$ पर स्थित होने जरूरी है
 (B) P तथा Q संगामी होंगे
 (C) P तथा Q को जोड़ने वाली रेखा का ढाल शून्य होगा
 (D) P तथा Q को जोड़ने वाली रेखा का ढाल -1 हो सकता है

Ans. ABC

Sol. Function and its inverse function may or may not intersect at line $y = x$.

D02/2/067. Which of the following is/are true :

- (A) Domain of $f(x) = \sqrt{\cos(\sin x)} + \sqrt{\sec(\sin x)} + \sqrt{x^2 - 4x + 10} + e^{\sqrt{\ln(x^2+1)}}$ is $\mathbb{R}$.
 (B) $f: A \rightarrow B$ where $n(A) = 5$ and $n(B) = 3$ then number of onto functions are 150
 (C) Period of $f(x) = e^{\left\{\frac{2x}{3}+5\right\}} + \cos^{10}(2\pi x)$ is $\frac{3}{2}$.
 (D) $f: (-1, 1) \rightarrow \mathbb{R}$, $f(x) = e^{\sec \tan \{x^3\}} [x^4] + \cos x$ is an even function

[where $[.]$ represents greatest integer function and $\{.\}$ represents fractional part function]

निम्न में से कौनसा/कौनसे कथन सत्य है :

- (A) $f(x) = \sqrt{\cos(\sin x)} + \sqrt{\sec(\sin x)} + \sqrt{x^2 - 4x + 10} + e^{\sqrt{\ln(x^2+1)}}$ का प्रान्त $\mathbb{R}$ है
 (B) $f: A \rightarrow B$ जहाँ $n(A) = 5$ तथा $n(B) = 3$ है तो आच्छादक फलनों की संख्या 150 होगी
 (C) $f(x) = e^{\left\{\frac{2x}{3}+5\right\}} + \cos^{10}(2\pi x)$ का आवर्तकाल $\frac{3}{2}$ है
 (D) $f: (-1, 1) \rightarrow \mathbb{R}$, $f(x) = e^{\sec \tan \{x^3\}} [x^4] + \cos x$ एक सम फलन है

[जहाँ $[.]$ महत्तम पूर्णांक फलन तथा $\{.\}$ भिन्नात्मक फलन को प्रदर्शित करता है]

Ans. ABCD

Sol. (A) $\cos(\sin x) \geq 0 \forall x \in \mathbb{R}$

$\sec(\sin x) \geq 0 \forall x \in \mathbb{R}$

$x^2 - 4x + 10 \Rightarrow D < 0 \Rightarrow x^2 - 4x + 10 \geq 0 \forall x \in \mathbb{R}$

$$\ln(x^2 + 1) \geq 0 \quad \forall x \in \mathbb{R} \quad \& \quad x^2 + 1 > 0 \quad \forall x \in \mathbb{R}$$

So domain is $\mathbb{R}$

(B) Divide 5 object into group of 3 and distribute

$$(i) 2, 2, 1 \Rightarrow \frac{5!}{3!2!1!2!} \times 3! = \frac{120 \times 6}{8} = 90$$

$$(ii) 3, 1, 1 \Rightarrow \frac{5!}{3!1!1!2!} \times 3! = \frac{120 \times 6}{12} = 60$$

Total onto functions are 150

$$(C) \text{ Period of } e^{\left\{\frac{2x}{3}+5\right\}} \text{ is } \frac{3}{2}$$

$$\text{Period of } \cos^{10}(2\pi x) \text{ is } \frac{\pi}{2\pi} = \frac{1}{2}$$

$$\text{Period} \Rightarrow \text{LCM} \left\{ \frac{3}{2}, \frac{1}{2} \right\} \text{ of is } \frac{3}{2}$$

$$(D) \text{ as } x \in (-1, 1) \Rightarrow [x^4] = 0$$

$$f(x) = e^{\sec \tan \{x^3\}}(0) + \cos x$$

$$f(x) = \cos x \Rightarrow \text{Even function}$$

D02/2/068. $f: \mathbb{R} \rightarrow [-1, 1]$, $f(x) = \sin\left(\frac{a\pi x}{x^2 + 1}\right)$ is a surjective function if 'a' belongs to the set :

$f: \mathbb{R} \rightarrow [-1, 1]$, $f(x) = \sin\left(\frac{a\pi x}{x^2 + 1}\right)$ एक आच्छादक फलन है तब 'a' निम्न में से किस समुच्चय में होगा :

$$(A) (-\infty, -1] \cup [1, \infty) \quad (B) (-1, 1)$$

$$(C) (-\infty, -1]$$

$$(D) (1, \infty)$$

Ans. ACD

Sol.

D02/2/069. $f(x) = \left(\log_e \left[x \left[\frac{1}{x} \right] \right] \right)$, where $[]$ represents greatest integer function :

$$(A) \text{ Domain of } f(x) \text{ is } (-\infty, 0) \cup \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\} \quad (B) \text{ Domain of } f(x) \text{ is } \left\{ \frac{1}{n} \mid n \in \mathbb{Z}, n \neq 0 \right\}$$

$$(C) \text{ Range of } f(x) \text{ is } \{\ln(n) \mid n \in \mathbb{N}\}$$

$$(D) \text{ Range of } f(x) \text{ contains exactly one integer}$$

$$f(x) = \left(\log_e \left[x \left[\frac{1}{x} \right] \right] \right), \text{ जहाँ } [] \text{ महत्तम पूर्णांक फलन को दर्शाता है :}$$

(A) $f(x)$ का प्रांत $(-\infty, 0) \cup \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\}$ है

(B) $f(x)$ का प्रांत $\left\{ \frac{1}{n} \mid n \in \mathbb{Z}, n \neq 0 \right\}$ है

(C) $f(x)$ का परिसर $\{\ln(n) \mid n \in \mathbb{N}\}$ है

(D) $f(x)$ का परिसर ठीक दो पूर्णांक रखता है

Ans. ACD

Sol. $\left[x \left[\frac{1}{x} \right] \right] > 0$

$$x \left[\frac{1}{x} \right] \geq 1$$

$$x \left(\frac{1}{x} - \left\{ \frac{1}{x} \right\} \right) \geq 1$$

$$x \left\{ \frac{1}{x} \right\} \leq 0$$

$$x \in (-\infty, 0) \cup \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\}$$

$$\left[x \left[\frac{1}{x} \right] \right] = \text{integer}$$

$$\Rightarrow f(x) \in \{\ln(n) \mid n \in \mathbb{N}\}$$

D02/2/070. $f(x) = |\cos(\sin x)| - |\cos(\cos x)|$, then :

(A) Range of $f(x)$ is $[\cos 1 - 1, 1 - \cos 1]$

(B) Fundamental period of $f(x)$ is π

(C) $f(x)$ is increasing in $\left[0, \frac{\pi}{2} \right]$

(D) $f(x)$ is increasing in $\left[\frac{\pi}{2}, \pi \right]$

$f(x) = |\cos(\sin x)| - |\cos(\cos x)|$, तब :

(A) $f(x)$ का परिसर $[\cos 1 - 1, 1 - \cos 1]$ है

(B) $f(x)$ का मूलभूत आवर्तकाल π है

(C) $f(x)$, $\left[0, \frac{\pi}{2} \right]$ में वर्धमान है

(D) $f(x)$, $\left[\frac{\pi}{2}, \pi \right]$ में वर्धमान है

Ans. ABD

Sol. $f(x) = \cos(\sin x) - \cos(\cos x)$

Fundamental period π

$$f'(x) = -[\sin(\sin x) \cdot \cos x + \sin(\cos x) \sin x]$$

$$f'(x) = \begin{cases} < 0 & \forall x \in \left(0, \frac{\pi}{2}\right) \\ > 0 & \forall x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$$

$$f(x) \in [\cos 1 - 1, 1 - \cos 1]$$

D02/2/071. $f: \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = ax + e^x$. Then :

- (A) If $a = 0$, $f(x)$ is bijective
- (B) If $a > 0$, $f(x)$ is bijective
- (C) If $a < 0$, $f(x)$ is neither surjective nor injective
- (D) If $a < 0$, $f(x)$ is non-monotonic

$f: \mathbb{R} \rightarrow \mathbb{R}$ तथा $f(x) = ax + e^x$ हैं। तब :

- (A) यदि $a = 0$, $f(x)$ एकैकी-आच्छादक होगा
- (B) यदि $a > 0$, $f(x)$ आच्छादक होगा
- (C) यदि $a < 0$, $f(x)$ ना तो एकैकी और ना ही आच्छादक होगा
- (D) यदि $a < 0$, $f(x)$ ना तो वर्धमान और ना ही ह्रासमान होगा

Ans. BCD

- Sol.**
- (A) $f(x) = e^x \Rightarrow$ Injective but not surjective
 - (B) $f'(x) = a + e^x > 0 \Rightarrow$ Bijective
 - (C) $f'(x) = a + e^x \Rightarrow$ Neither injective not surjective
 - (D) $a < 0 \Rightarrow f(x)$ is non-monotonic

D02/2/072. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \left\{ \ln(\sqrt{1+x^2} + x) - \ln(\sqrt{1+x^2} - x) \right\}^3$, then :

(A) $f^{-1}(x) = \frac{e^{\sqrt[3]{x}} - e^{-\sqrt[3]{x}}}{2}$ (B) Range of $f(x)$ is $(-\infty, \infty)$

(C) $f(x)$ is an increasing function (D) $f^{-1}(x) = \frac{e^{\frac{\sqrt{x}}{2}} - e^{-\frac{\sqrt{x}}{2}}}{2}$

माना $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \left\{ \ln(\sqrt{1+x^2} + x) - \ln(\sqrt{1+x^2} - x) \right\}^3$, तब :

(A) $f^{-1}(x) = \frac{e^{\sqrt[3]{x}} - e^{-\sqrt[3]{x}}}{2}$ (B) $f(x)$ का परिसर $(-\infty, \infty)$ है

(C) $f(x)$ एक वर्धमान फलन है

$$(D) f^{-1}(x) = \frac{e^{\frac{\sqrt[3]{x}}{2}} - e^{-\frac{\sqrt[3]{x}}{2}}}{2}$$

Ans. BCD

Sol.
$$f(x) = \left(\ln \frac{\sqrt{1+x^2} - x}{\sqrt{1+x^2} + x} \right)^3$$

$$f(x) = 8 \left(\ln x + \sqrt{1+x^2} \right)^3$$

$$f'(x) = \frac{24 \left(\ln x + \sqrt{1+x^2} \right)^2}{\sqrt{1+x^2}} > 0$$

$$\Rightarrow y = 8 \left(\ln x + \sqrt{1+x^2} \right)^3$$

$$e^{\frac{\sqrt[3]{y}}{2}} = x + \sqrt{1+x^2}$$

$$e^{\frac{\sqrt[3]{y}}{2}} = 1 + 2x^2 + 2x \left(e^{\frac{\sqrt[3]{y}}{2}} - x \right)$$

$$x = \frac{e^{\frac{\sqrt[3]{y}}{2}} - e^{-\frac{\sqrt[3]{y}}{2}}}{2}$$

D02/2/073. If $f(x) = \frac{1}{\left[[x-1] \right] + \left[[7-x] \right] - 6}$ then set of values of x which not belongs to the domain is(are) :

(A) $x \in (0, 1]$ (B) $x \in [7, 8)$ (C) $x \in \{2, 3, 4\}$ (D) $x \in (7, 10)$

Note : $[x]$ denote greatest integer of x respectively.

यदि $f(x) = \frac{1}{\left[[x-1] \right] + \left[[7-x] \right] - 6}$ तब x के उन मानों का समुच्चय जो प्रांत में नहीं है/हैं :

(A) $x \in (0, 1]$ (B) $x \in [7, 8)$ (C) $x \in \{2, 3, 4\}$ (D) $x \in (7, 10)$

नोट : $[x]$, x के सापेक्ष महत्तम पूर्णांक को दर्शाता है।

Ans. ABC

Sol.
$$f(x) = \frac{1}{\left[[x-1] \right] + \left[[7-x] \right] - 6}$$

$$[x-1] + [7-x] - 6 \neq 0$$

Case 1 : $1 < x < 7$

$$[x-1] + [7-x] - 6 \neq 0$$

$$[x] - 1 + [-x] + 7 - 6 \neq 0$$

$$[x] + [-x] \neq 0$$

$$x \notin I$$

$$x \in (1, 7) - \{2, 3, 4, 5, 6\}$$

Case 2 : $x \leq 1$

$$[1-x] + [7+x] - 6 \neq 0$$

$$2 + 2[-x] \neq 0$$

$$[-x] \neq -1$$

$$-x \notin [-1, 0)$$

$$\Rightarrow x(0, 1] \Rightarrow x \in (-\infty, 0]$$

Case 3 : $[x-1] + [x-7] - 6 \neq 0$

$$2[x] \neq 14$$

$$[x] \neq 7$$

$$x \notin [7, 8)$$

Using can I, II and III we get :

$$x \in \mathbb{R} - (0, 1] - [7, 8) - \{2, 3, 4, 5, 6\}$$

D02/2/074. Let $f: \mathbb{R} \rightarrow [1, \infty)$ be a quadratic surjective function such that $f(2+x) = f(2-x)$ and $f(1) = 2$. If $g: (-\infty, \ln 2] \rightarrow [1, 5)$ is given by $g(\ln x) = f(x)$ then which of the following is(are) correct ?

(A) The value of $f(3)$ is equal to 2

(B) $g^{-1}(x) = \ln(2 - \sqrt{x-1})$

(C) $g^{-1}(x) = \ln(2 + \sqrt{x-1})$

(D) The sum of values of x satisfying the equation $f(x) = 5$ is 4

माना $f: \mathbb{R} \rightarrow [1, \infty)$ में एक द्विघातीय आच्छादक फलन इस प्रकार है कि $f(2+x) = f(2-x)$ तथा $f(1) = 2$ है। यदि $g: (-\infty, \ln 2] \rightarrow [1, 5)$, $g(\ln x) = f(x)$ द्वारा दिया जाता है तब निम्न में से कौनसा/से सत्य होगा/होंगे ?

(A) $f(3)$ का मान 2 के बराबर है

(B) $g^{-1}(x) = \ln(2 - \sqrt{x-1})$

(C) $g^{-1}(x) = \ln(2 + \sqrt{x-1})$

(D) समीकरण $f(x) = 5$ को संतुष्ट करने वाले x के मानों का योग 4 है

Ans. ABD

Sol. $f(x) = a(x-2)^2 + 1$

$$\therefore f(1) = 2 \quad \therefore a = 1$$

$$f(x) = x^2 - 4x + 5$$

$$g(\ln x) = x^2 - 4x + 5 = (x - 2)^2 + 1$$

$$g(x) = (e^x - 2)^2 + 1$$

$$x \in (-\infty, \ln 2) \Rightarrow g(x) \in [1, 5)$$

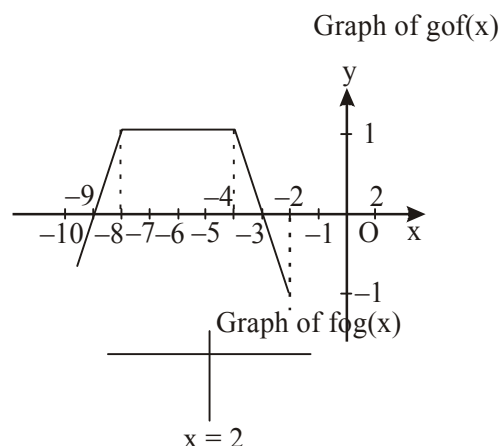
$\Rightarrow g(x)$ is invertible function.

$$y = (e^x - 2)^2 + 1$$

$$e^x = 2 + \sqrt{y-1}, 2 - \sqrt{y-1}$$

$$x = \ln(2 - \sqrt{y-1}) \text{ as } x \in (-\infty, \ln 2]$$

$$g^{-1}(x) = \ln(2 - \sqrt{x-1})$$



D02/2/075. Which of the following statement(s) is(are) correct?

- (A) If f is a one-one mapping from set A to A , then f is onto.
- (B) If f is an onto mapping from set A to A , then f is one-one.
- (C) Let f and g be two functions defined from $\mathbb{R} \rightarrow \mathbb{R}$ such that $g \circ f$ is injective, then f must be injective.
- (D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.

निम्न में से कौनसा/से कथन सत्य है/हैं?

- (A) यदि f , समुच्चय A से A में एकैकी प्रतिचित्रण है, तब f आच्छादक है।
- (B) यदि f , A से A में आच्छादक प्रतिचित्रण है तब f एकैकी है।
- (C) माना f तथा g दो फलन $\mathbb{R} \rightarrow \mathbb{R}$ में इस प्रकार परिभाषित हैं कि $g \circ f$ एकैकी है, तब f एकैकी होगा।
- (D) यदि समुच्चय A में 3 अवयव हैं जबकि समुच्चय B में 2 अवयव हैं, तब A से B में कुल फलों की संख्या 8 है।

Ans. CD

Sol. (A) $f: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$f(x) = \sin x \Rightarrow 1 - 1 \text{ but not onto}$$

$$(B) f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = x^2 \Rightarrow \text{onto but not one-one}$$

D02/2/076. If $f(x) = x^3 + x^2 + 2x$, $g(x) = e^x$ (where $g(x)$ is invertible function), then which of the following statement(s) is/are correct?

- (A) $g \circ f(x) \leq g(x)$ for $x \in (-\infty, 0]$
- (B) $g^{-1} \circ f(x) \geq g^{-1}(x)$ for $x \in (0, \infty)$
- (C) $g^{-1} \circ f(x) \leq g^{-1}(x)$ for $x \in (0, \infty)$
- (D) $g^{-1} \circ f(x) \geq g^{-1}(x)$ for $x \in (-\infty, 0]$

यदि $f(x) = x^3 + x^2 + 2x$, $g(x) = e^x$ (जहाँ $g(x)$ व्युत्क्रमणीय फलन है), तब निम्न में से कौनसा/से कथन सत्य होगा/होंगे?

- (A) $x \in (-\infty, 0]$ के लिए $g \circ f(x) \leq g(x)$
- (B) $x \in (0, \infty)$ के लिए $g^{-1} \circ f(x) \geq g^{-1}(x)$
- (C) $x \in (0, \infty)$ के लिए $g^{-1} \circ f(x) \leq g^{-1}(x)$
- (D) $x \in (0, \infty)$ के लिए $g^{-1} \circ f(x) \geq g^{-1}(x)$

Ans. AB

Sol. (A) $e^{x^3+x^2+2x} \leq e^x$

$$x^3 + x^2 + 2x \leq x$$

$$(B) \ln^{(x^3+x^2+x)} \geq \ln^x \quad x > 0$$

$$x^3 + x^2 + 2x \leq 0 \quad x > 0$$

$$x \in (0, \infty)$$

D02/2/077. Consider the following sets

$$A = \left\{ \frac{x^2 - 2x + 4}{x^2 + 2x + 4} \in \mathbb{R} : x \in \mathbb{R} \right\},$$

$$B = \left\{ \frac{x^2 - 2x - 4}{x^2 + 2x - 4} \in \mathbb{R} : x \in \mathbb{R} \right\}.$$

Then which of the following options is/are correct?

समुच्चयों

$$A = \left\{ \frac{x^2 - 2x + 4}{x^2 + 2x + 4} \in \mathbb{R} : x \in \mathbb{R} \right\},$$

$$B = \left\{ \frac{x^2 - 2x - 4}{x^2 + 2x - 4} \in \mathbb{R} : x \in \mathbb{R} \right\}.$$

तब निम्न में से कौनसा/से विकल्प सही होगा/होंगे?

$$(A) \left\{ 1, \frac{3}{7}, \frac{7}{3}, 3, \frac{1}{3}, \frac{7}{19}, \frac{19}{7} \right\} \subseteq A$$

$$(B) \{-13, 2301, \pi, -e\} \subseteq B$$

$$(C) \left\{ 1, \frac{3}{7}, \frac{7}{3}, 3, \frac{1}{3}, \frac{19}{39}, \frac{39}{19} \right\} \subseteq A$$

$$(D) \{-1301, 23, -\pi, e\} \subseteq B$$

Ans. ABCD

Sol. $A \in \left[\frac{1}{3}, 3 \right]$

$$B \in \left(-\infty, -3 - 2\sqrt{3} \right] \cup \left[-3 + 2\sqrt{3}, \infty \right)$$

D02/2/078. Which of the following options is/are correct, if

$$f(x) = \max(x^2, |2 - x|) \text{ and } g(x) = \min(x^3, 3x - 2)$$

$$(A) \text{ If } x \in (-\infty, -2], \text{ then } f(g(x)) = x^6$$

$$(B) \text{ If } x \in (-2, 0], \text{ then } f(g(x)) = 9x^2 - 12x + 4$$

$$(C) \text{ If } x \in (0, 1], \text{ then } f(g(x)) = 4 - 3x$$

$$(D) \text{ If } x \in (-2, 1], \text{ then } g(f(x)) = 4 - 3x$$

निम्न में से कौनसा/से विकल्प सत्य है/हैं, यदि

$$f(x) = \max(x^2, |2 - x|) \text{ तथा } g(x) = \min(x^3, 3x - 2)$$

$$(A) \text{ यदि } x \in (-\infty, -2], \text{ तब } f(g(x)) = x^6$$

$$(B) \text{ यदि } x \in (-2, 0], \text{ तब } f(g(x)) = 9x^2 - 12x + 4$$

$$(C) \text{ यदि } x \in (0, 1], \text{ तब } f(g(x)) = 4 - 3x$$

$$(D) \text{ यदि } x \in (-2, 1], \text{ तब } g(f(x)) = 4 - 3x$$

Ans. ACD

Sol.

$$f(x) = \begin{cases} x^2 & x < -2 \\ 2 - x & -2 \leq x \leq 1 \\ x^2 & 1 < x \end{cases}$$

$$g(x) = \begin{cases} x^3 & x < -2 \\ 3x - 2 & -2 \leq x \end{cases}$$

$$f(g(x)) = \begin{cases} x^6 & x < -2 \\ (3x - 2)^2 & -2 \leq x \leq 0 \\ 4 - 3x & 0 < x \leq 1 \\ (3x - 2)^2 & 1 < x \end{cases}$$

$$g(f(x)) = \begin{cases} 3x^2 - 2 & x < -2 \\ 4 - 3x & -2 \leq x \leq 1 \\ 3x^2 - 2 & 1 < x \end{cases}$$

D02/2/079. Let $a, b, c \in \mathbb{R}$, $a + b + c = 3$ and functions f, g, h be such that

$$f: \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]; f(x) = \frac{x}{1+x^2},$$

$$g(x) = ax^2 + bx + c; g(x+y) = g(x) + g(y) + xy \quad \forall x, y \in \mathbb{R} \text{ and}$$

$$h(x) + 2h\left(\frac{1}{x}\right) = 3x \quad \forall x \in \mathbb{R} - \{0\}.$$

If $|X|$ denotes the cardinal number of set X , then which of the following options is/are correct?

$$(A) f \text{ is Injective function}$$

$$(B) f \text{ is Surjective function}$$

$$(C) g(1) + g(2) + g(3) + \dots + g(10) = 330$$

$$(D) |\{x \in \mathbb{R} : h(x) = h(-x)\}| = 2$$

माना $a, b, c \in \mathbb{R}$, $a + b + c = 3$ तथा फलन f, g, h इस प्रकार हैं कि

$$f: \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]; f(x) = \frac{x}{1+x^2},$$

$$g(x) = ax^2 + bx + c; g(x+y) = g(x) + g(y) + xy \quad \forall x, y \in \mathbb{R} \text{ तथा}$$

$$h(x) + 2h\left(\frac{1}{x}\right) = 3x \quad \forall x \in \mathbb{R} - \{0\}.$$

यदि $|X|$ समुच्चय X में आने वाले अवयवों की संख्या को दर्शाता है तब निम्न में से कौनसा/से कथन सत्य है/हैं?

(A) f एकैकी फलन है

(B) f आच्छादक फलन है

(C) $g(1) + g(2) + g(3) + \dots + g(10) = 330$

(D) $|\{x \in \mathbb{R} : h(x) = h(-x)\}| = 2$

Ans. BCD

Sol. $f: \mathbb{R} \rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$

(A), (B) $f(x) = \frac{x}{1+x^2}$

$$f'(x) = \frac{(1+x^2) - 2x^2}{(1+x^2)^2} = \frac{1-x^2}{1+x^2}$$

$f(x)$ is not injective.

$$f(x) = \frac{1}{x + \frac{1}{x}} \Rightarrow f(x) \leftarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$$

(C) $g(x) = ax^2 + bx + c$

$$g(x+y) = g(x) + g(y) + xy$$

put $y = 0$

$$g(x) = g(x) + c$$

$$c = 0$$

$$x = y = 1$$

$$g(2) = 2g(1) + 1$$

$$4a + 2b = 2a + 2b + 1$$

$$2a = 1 \Rightarrow a = \frac{1}{2}$$

$$\therefore a + b + c = 3$$

$$b = \frac{5}{2}$$

$$\begin{aligned}\sum_{x=1}^{10} g(x) &= \sum_{x=1}^{10} \left(\frac{x^2}{2} + \frac{5x}{2} \right) \\ &= \frac{1}{2} \times \frac{10 \times 11 \times 21}{6} + \frac{5}{2} \times \frac{10 \times 11}{2} \\ &= 330\end{aligned}$$

$$(D) \quad h(x) + 2h\left(\frac{1}{x}\right) = 3x \quad \dots(1)$$

$$2h(x) + h\left(\frac{1}{x}\right) = \frac{3}{x} \quad \dots(2)$$

Solving (1) & (2)

$$h(x) = \frac{2}{x} - x$$

$$h(x) = h(-x)$$

$$\frac{2}{x} - x = -\frac{2}{x} + x$$

$$4 = x^2$$

$$x = \pm 2$$

$$|x \in \mathbb{R} : h(x) = h(-x)| = 2$$

D03/2/080. Which of the following option(s) lie in the range of

$$f(x) = \frac{818 \left(\cot^{-1} x + \sec^{-1} x + \csc^{-1} x \right)}{\sin \left(\frac{1}{2} \cdot \cos^{-1} \frac{1}{2} \right) + \sin \left(2 \cdot \cos^{-1} \frac{5}{13} \right)} :$$

$$(A) \quad 339\pi$$

$$(B) \quad 507\pi$$

$$(C) \quad 845\pi$$

$$(D) \quad 1014\pi$$

Ans. ABC

Sol.

D03/2/081. Which of the following options is/are correct for $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \cos^{-1}(-\{x\}); \{x\} = x - \max \{a \in \mathbb{Z} : a \leq x\} \text{ and}$$

$$g(x) = \cos^{-1}(\cot^{-1} \ln x) :$$

$$(A) \quad \text{Domain of } g \text{ is } [e^{\cot^{-1}}, \infty) \text{ and range is } \left[0, \frac{\pi}{2} \right]$$

$$(B) \quad f \text{ is neither an Injective function nor an Even function}$$

(C) Range of f contains two prime numbers

(D) f is periodic function with period 1

Ans. BCD

Sol.

D03/2/082. Which of the following options is/are correct, if set A defined below contains 4 elements

$$A = \left\{ x \in \mathbb{R} : \left| \tan^{-1} 3x \right| = \frac{3}{4}|x| + \frac{\pi}{3} - \frac{\sqrt{3}}{k} \right\} :$$

(A) $k = 1$

(B) $k = 2$

(C) $k = 3$

(D) $k = 4$

Ans. BC

Sol.

D03/2/083. If $\lfloor x \rfloor = \max \{ a \in \mathbb{Z} : a \leq x \}$, then which of the following option(s) lie in set

$$\left\{ \left(4 \tan^{-1} \sqrt{\lfloor x \rfloor + \lfloor -x \rfloor} + 4\sqrt{2 - |x|} + \frac{4}{x^2}, 2 \sin^{-1} \lfloor 2y \rfloor + 3 \cos^{-1} (\lfloor y \rfloor - 1) \right) \in \mathbb{R}^2 : (x, y) \in \mathbb{R}^2 \right\} :$$

(A) $(1, 3\pi)$

(B) $(8, 4\pi)$

(C) $(4, 4\pi)$

(D) $(12, 5\pi)$

Ans. AB

Sol.

D03/2/084. Let function f and g be defined as

$$f : \left[-\frac{\pi}{3}, \frac{\pi}{6} \right] \rightarrow [0, 4]; f(x) = 2 \cos^2 x + \sqrt{3} \sin 2x + 1 \text{ and}$$

$$g(x) = \min \left(\tan^{-1}(x+2), \cot^{-1}(x+2) \right) - \frac{\pi}{3}.$$

If $\alpha < \beta$ are the integers lying in the range of g , then :

(A) $4f^{-1}(1) > -\pi$

(B) $\cos^{-1} \cos \alpha + \sin^{-1} \sin \beta = 1$

(C) $f^{-1}(\sqrt{1.25} + 2.5) = \frac{\pi}{15}$

(D) $\cos^{-1} \cos \beta + \sin^{-1} \sin \alpha = 3 - \pi$

Ans. ABCD

Sol.

D03/2/085. Which of the following options is/are correct, if

$$x_1 = \sin\left(\frac{1}{3}\sin^{-1}\frac{1}{\sqrt{3}}\right), \quad x_2 = \cos\left(\cos^{-1}\frac{1}{\sqrt{5}} - \sin^{-1}\frac{2}{\sqrt{5}}\right),$$

$$\{\sin \cot^{-1} \cos \tan^{-1} x \in \mathbb{R} : x \in \mathbb{R}\} = [p, q]; p, q \in \mathbb{R}.$$

Here $\lfloor x \rfloor = \max \{a \in \mathbb{Z} : a \leq x\}$:

$$(A) \quad x_2^2 = 3(3x_1 - 4x_1^3)^2$$

$$(B) \quad 2x_2^2 = 3(3x_1 - 4x_1^3)^2$$

$$(C) \quad \left\lfloor q + \frac{1}{p} \right\rfloor = 2$$

$$(D) \quad \left\lfloor q + \frac{1}{p} \right\rfloor = 3$$

Ans. AC

Sol.

D03/2/086. Let $|X|$ denote the cardinal number of set X ,

$$A = \left\{ x \in \mathbb{R} : 2 \tan^{-1} |x| = \frac{1}{\ln |x|} \right\},$$

$$\sec^{-1} \sqrt{1+x^2} + \csc^{-1} \frac{\sqrt{1+y^2}}{y} + \cot^{-1} \frac{1}{z} = \pi; x, y, z > 0 \text{ and}$$

$$\tan^{-1}(a+1) + \cot^{-1}(a-1) = \sin^{-1} \frac{4}{5} + \cos^{-1} \frac{3}{5}.$$

Then, which of the following options is/are correct?

$$(A) \quad |A| = 2$$

$$(B) \quad x + y + z = xyz$$

$$(C) \quad 7a^2 = 48$$

$$(D) \quad |A| > 2$$

Ans. ABC

Sol.

D04/2/087. Let $f(x) = \frac{\sin^2(mx)}{nx^2}$; $x \neq 0$ where $m, n \in \{1, 2, 3, 4, 5\}$ & $L = \lim_{x \rightarrow 0} f(x)$ then :

(A) Number of possible values of L is 12 if $m+n$ is odd

(B) Number of possible values of L is 9 if $m+n$ is odd

(C) Number of possible integral values of L is 9

(D) Number of possible integral values of L is 5

माना $f(x) = \frac{\sin^2(mx)}{nx^2}$; $x \neq 0$ जहाँ $m, n \in \{1, 2, 3, 4, 5\}$ तथा $L = \lim_{x \rightarrow 0} f(x)$ तब :

- (A) L के संभावित मानों की संख्या 12 है यदि $m + n$ विषम है
 (B) L के संभावित मानों की संख्या 9 है यदि $m + n$ विषम है
 (C) L के संभावित पूर्णांक मानों की संख्या 9 है
 (D) L के संभावित पूर्णांक मानों की संख्या 5 है

Ans. AC

Sol. $L = \frac{m^2}{n}$; $m, n \in \{1, 2, 3, 4, 5\}$

L is integer when

$$n = 1; m = 1, 2, 3, 4, 5$$

$$n = 2; m = 2, 4$$

$$n = 3; m = 3$$

$$n = 4; m = 4$$

$$n = 5; m = 5$$

$$\therefore L = 1, 4, 9, 16, 25, 28, 35 \quad (C)$$

$$m + n = \text{odd when one of them is even \& other is odd} \quad (A)$$

D04/2/088. If $\lim_{x \rightarrow 0} \sin\left(\frac{\pi(1 - \cos^m x)}{x^n}\right)$ exist $\forall m \in \mathbb{N}$ then :

यदि $\lim_{x \rightarrow 0} \sin\left(\frac{\pi(1 - \cos^m x)}{x^n}\right)$ $\forall m \in \mathbb{N}$ के लिए विद्यमान है तब :

$$(A) n = 1$$

$$(B) n = 2$$

$$(C) n = 3$$

$$(D) n \in \mathbb{N}$$

Ans. AB

Sol. $\lim_{x \rightarrow 0} \sin\left(\frac{\pi(1 - \cos^m x)}{x^n}\right) = \text{exist} = \sin\left(\lim_{x \rightarrow 0} \frac{\pi(1 - \cos^m x)}{x^n}\right) = \sin\left(\lim_{x \rightarrow 0} \frac{\pi \cdot m \sin x \cos^{m-1} x}{n \cdot x^{n-1}}\right)$

D04/2/089. Let $f(x) = \frac{3}{4}x + 1$, $f^n(x)$ be defined as $f^2(x) = f(f(x))$, and for $n \geq 2$, $f^{n+1}(x) = f(f^n(x))$. If $\lambda = \lim_{n \rightarrow \infty} f^n(x)$,

then :

(A) λ is independent of x

(B) λ is a linear polynomial in x

(C) the line $y = \lambda$ has slope 0

(D) the line $4y = \lambda$ touches the unit circle with center at the origin

माना $f(x) = \frac{3}{4}x + 1$ है, $f^n(x)$ इस प्रकार प्रदर्शित है कि $f^2(x) = f(f(x))$, तथा $n \geq 2$ के लिए $f^{n+1}(x) = f(f^n(x))$ है। यदि

$\lambda = \lim_{n \rightarrow \infty} f^n(x)$ है, तो:

(A) λ, x से स्वतंत्र है

(B) λ, x में एक रेखीय बहुपद है

(C) रेखा $y = \lambda$ का ढाल 0 है

(D) रेखा $4y = \lambda$ उस वृत्त को स्पर्श करती है जिसका केन्द्र मूल बिन्दु पर है और त्रिज्या 1 है।

Ans. ACD

Sol.

$$f^2(x) = f\left(\frac{3}{4}x + 1\right) = \frac{3}{4}\left(\frac{3}{4}x + 1\right) + 1 = \left(\frac{3}{4}\right)^2 x + \frac{3}{4} + 1$$

$$f^3(x) = f\{f^2(x)\} = \frac{3}{4}\{f^2(x) + 1\}$$

$$= \frac{3}{4}\left\{\left(\frac{3}{4}\right)^2 x + \frac{3}{4} + 1\right\} + 1$$

$$= \left(\frac{3}{4}\right)^3 x + \left(\frac{3}{4}\right)^2 + \frac{3}{4} + 1$$

$$\therefore f^n(x) = \left(\frac{3}{4}\right)^n x + \left(\frac{3}{4}\right)^{n+1} + \left(\frac{3}{4}\right)^{n+2} + \dots + \left(\frac{3}{4}\right) + 1$$

$$= \left(\frac{3}{4}\right)^n x + \frac{1 - \left(\frac{3}{4}\right)^n}{1 - \frac{3}{4}}$$

$$\therefore \lim_{n \rightarrow \infty} f^n(x) = 0 + 4 = 4$$

D04/2/090. If $f(\alpha) = \lim_{x \rightarrow 2} (\sin^x \alpha + \cos^x \alpha)^{\frac{1}{(x-2)}}$ for $\alpha \in \left[0, \frac{\pi}{2}\right]$, then :

(A) $f(0) = 1$

(B) $f\left(\frac{\pi}{2}\right) = 1$

$$(C) f(\alpha) = (\cos \alpha)^{\cos^2 \alpha} \cdot (\sin \alpha)^{\sin^2 \alpha} \text{ if } \alpha \in \left(0, \frac{\pi}{2}\right)$$

$$(D) f(\alpha) = \frac{(\sin \alpha)^{\sin^2 \alpha}}{(\cos \alpha)^{\cos^2 \alpha}} \text{ if } \alpha \in \left(0, \frac{\pi}{2}\right)$$

$$\text{यदि } f(\alpha) = \lim_{x \rightarrow 2} (\sin^x \alpha + \cos^x \alpha)^{\frac{1}{(x-2)}} \forall \alpha \in \left[0, \frac{\pi}{2}\right], \text{ तब :}$$

$$(A) f(0) = 1$$

$$(B) f\left(\frac{\pi}{2}\right) = 1$$

$$(C) f(\alpha) = (\cos \alpha)^{\cos^2 \alpha} \cdot (\sin \alpha)^{\sin^2 \alpha} \text{ है यदि } \alpha \in \left(0, \frac{\pi}{2}\right)$$

$$(D) f(\alpha) = \frac{(\sin \alpha)^{\sin^2 \alpha}}{(\cos \alpha)^{\cos^2 \alpha}} \text{ है यदि } \alpha \in \left(0, \frac{\pi}{2}\right)$$

Ans. ABC

$$\text{Sol. } f(\alpha) = \lim_{x \rightarrow 2} (\sin^x \alpha + \cos^x \alpha)^{\frac{1}{(x-2)}} \quad (1^\infty \text{ form})$$

$$= \begin{cases} e^{\lim_{x \rightarrow 2} \frac{\sin^x \alpha + \cos^x \alpha - 1}{x - 2}} & , \quad \alpha \in \left(0, \frac{\pi}{2}\right) \\ 1 & , \quad \alpha = 0, \frac{\pi}{2} \end{cases}$$

$$\text{Now, } e^{\lim_{x \rightarrow 2} \frac{\sin^x \alpha + \cos^x \alpha - 1}{x - 2}} = e^{\lim_{x \rightarrow 2} \frac{\sin^x \alpha + \cos^x \alpha - \sin^2 \alpha - \cos^2 \alpha}{x - 2}}$$

$$= e^{\lim_{x \rightarrow 2} \frac{\sin^2 \alpha (\sin^{x-2} \alpha - 1) + \cos^2 \alpha (\cos^{x-2} \alpha - 1)}{x - 2}}$$

$$= e^{\sin^2 \alpha \log_e \sin \alpha + \cos^2 \alpha \log_e \cos \alpha}$$

$$= e^{\log_e (\sin \alpha) \sin^2 \alpha + \log_e (\cos \alpha) \cos^2 \alpha}$$

$$= (\sin \alpha)^{\sin^2 \alpha} (\cos \alpha)^{\cos^2 \alpha}$$

$$\therefore f(x) = \begin{cases} (\cos \alpha) \cos^2 \alpha (\sin \alpha)^{\sin^2 \alpha}, & \alpha \in \left(0, \frac{\pi}{2}\right) \\ 1, & \alpha = 0, \frac{\pi}{2} \end{cases}$$

D04/2/091. Let $\lambda = \lim_{x \rightarrow 0} \left(\frac{\tan x \cdot \cos \sin x - \sin \sin x}{\tan x \cdot \cos \tan x - \sin \tan x} \right)$, $\mu = \lim_{x \rightarrow 0} \left(\frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4} \right)$; $a > 0$.

If μ is finite, then :

माना $\lambda = \lim_{x \rightarrow 0} \left(\frac{\tan x \cdot \cos \sin x - \sin \sin x}{\tan x \cdot \cos \tan x - \sin \tan x} \right)$, $\mu = \lim_{x \rightarrow 0} \left(\frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4} \right)$; $a > 0$ हैं।

यदि μ परिमित है, तब :

(A) $a - 2\lambda = 3$ (B) $64\mu + a = 3$ (C) $a - 2\lambda = 2$ (D) $32\mu = 1$

Ans. AB

Sol.
$$\frac{\sin x \cos(\sin x) - \cos x \sin(\sin x)}{\sin x \cos(\tan x) - \cos x \sin(\tan x)}$$

$$\lambda = \lim_{x \rightarrow 0} \frac{\sin(x - \sin x)}{\sin(x - \tan x)}$$

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{1 - \sec^2 x}$$

$$\lim_{x \rightarrow 0} \frac{+ \sin x}{-2 \sec x \times \sec x \tan x} = \lim_{x \rightarrow 0} \frac{x}{2 \sec^2 x \times x}$$

$$\lambda = -\frac{1}{2}$$

$$\mu = \lim_{x \rightarrow 0} \frac{a - \sqrt{a^2 - x^2} - \frac{x^2}{4}}{x^4}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{a - a \left(1 - \left(\frac{x}{a} \right)^2 \right)^{\frac{1}{2}} - \frac{x^2}{4}}{x^4}$$

$$\Rightarrow \frac{a - a \left(1 - \frac{1}{2} \frac{x^2}{a^2} - \frac{\frac{1}{2} \times \frac{1}{2} \times \frac{x^4}{a^4}}{2!} + \dots \right) - \frac{x^2}{4}}{x^4}$$

$$\Rightarrow \frac{a - a + \frac{x^2}{2a} - \frac{3}{8} \frac{x^4}{a^3} - \frac{x^2}{4}}{x^4} = \text{finite}$$

$$\frac{x^2}{2a} = -\frac{x^2}{4} \rightarrow a = 2$$

$$\mu = -\frac{1}{8} \times \frac{1}{a^3} = -\frac{1}{64}$$

$$64\mu = -1$$

D04/2/092. Let $\mu_1 = \lim_{a \rightarrow 0^+} (p)$, $\mu_2 = \lim_{a \rightarrow 0^+} (q)$, where p, q are roots of

$$\left(\sqrt[3]{1+a} - 1 \right) x^2 + \left(\sqrt{1+a} - 1 \right) x + \left(\sqrt[6]{1+a} - 1 \right) = 0 ; p > -1, \text{ and}$$

$$\lambda_1 = \lim_{x \rightarrow \infty} g(x), \lambda_2 = \lim_{x \rightarrow -\infty} g(x); \text{ where } g(x) = \frac{\ln(x^2 + e^x)}{\ln(x^4 + e^{2x})},$$

If $\mu_1 > \mu_2$, then which of the following options is/are correct ?

$$\text{माना } \mu_1 = \lim_{a \rightarrow 0^+} (p), \mu_2 = \lim_{a \rightarrow 0^+} (q), \text{ जहाँ } p, q, \text{ समीकरण } \left(\sqrt[3]{1+a} - 1 \right) x^2 + \left(\sqrt{1+a} - 1 \right) x + \left(\sqrt[6]{1+a} - 1 \right) = 0 ; p$$

$$> -1 \text{ के मूल हैं तथा } \lambda_1 = \lim_{x \rightarrow \infty} g(x), \lambda_2 = \lim_{x \rightarrow -\infty} g(x); \text{ जहाँ } g(x) = \frac{\ln(x^2 + e^x)}{\ln(x^4 + e^{2x})},$$

यदि $\mu_1 > \mu_2$, तब निम्न में से कौनसा/से विलय सही है/हैं ?

$$(A) 2\lambda_1 - \mu_2 = 2 \quad (B) 4\mu_2 - 2\mu_1 = 3 \quad (C) 3\lambda_1 - \mu_1 = 2 \quad (D) 3\lambda_2 - 5\mu_2 = 3$$

Ans. ABC

Sol. $(\sqrt[3]{1+a} - 1)x^2 + (\sqrt{1+a} - 1)x + (\sqrt[6]{1+a} - 1) = 0$

$$p + q = \frac{1 - \sqrt{1+a}}{(1+a)^{\frac{1}{3}} - 1}$$

taking $\lim_{a \rightarrow 0^+}$ both side

$$\lim_{a \rightarrow 0^+} p + \lim_{a \rightarrow 0^+} q = \lim_{a \rightarrow 0^+} \frac{1 - (1+a)^{\frac{1}{2}}}{(1+a)^{\frac{1}{3}} - 1}$$

$$\mu_1 + \mu_2 = \lim_{a \rightarrow 0^+} \frac{-\frac{1}{2}(1+a)^{-\frac{1}{2}}}{\frac{1}{3}(1+a)^{-\frac{2}{3}}}$$

$$\mu_1 + \mu_2 = -\frac{3}{2} = -1 - \frac{1}{2}$$

$$pq = \frac{(1+a)^{\frac{1}{6}} - 1}{(1+a)^{\frac{1}{3}} - 1}$$

taking $\lim_{a \rightarrow 0^+}$ both side

$$\mu_1 \mu_2 = \lim_{a \rightarrow 0^+} \frac{(1+a)^{\frac{1}{6}} - 1}{(1+a)^{\frac{1}{3}} - 1}$$

$$\mu_1 \mu_2 = \frac{\frac{1}{6}}{\frac{1}{3}} \Rightarrow \frac{1}{2}$$

$$\mu_1 = -\frac{1}{2}, \mu_2 = -1$$

$$g(x) = \frac{\ln x^2 + \ln \left(1 + \frac{e^x}{x^2}\right)}{\ln x^4 + \ln \left(1 + \frac{e^{2x}}{x^4}\right)}$$

$$\lambda_1 = \lim_{x \rightarrow \infty} \frac{1}{2}$$

$$\lambda_2 = \lim_{x \rightarrow -\infty} \frac{\ln(x^2 + e^x)}{\ln(x^4 + e^{2x})} \quad \text{put } x = -y$$

$$\lambda_2 = \lim_{x \rightarrow -\infty} \frac{\ln(y^2 e^y + 1) - y}{\ln(e^{2y} y^4 + 1) - 2y}$$

$$\Rightarrow \frac{\ln(e^y \cdot y^2) - y}{\ln(e^{2y} y^4) - 2y}$$

$$\lambda_2 = \frac{y + 2 \ln y - y}{2y + 4 \ln y - 2y} = \frac{1}{2}$$

$$\lambda_1 = \frac{1}{2}, \lambda_2 = \frac{1}{2}, \mu_1 = -\frac{1}{2}, \mu_2 = -1$$

D04/2/093. Let functions f and g are defined as

$$f(x) = \lim_{h \rightarrow 0} \frac{\sin(x+3h) - 3\sin(x+2h) + 3\sin(x+h) - \sin x}{h(\tan(x+2h) - 2\tan(x+h) + \tan x)} \text{ and}$$

$$g(x) = \frac{1-x(1+|1-x|)}{|1-x|} f\left(\frac{1}{1-x}\right) \sin \frac{1}{1-x} \sec^3 \frac{1}{1-x}.$$

Then which of the following options is/are correct ?

- (A) $\lim_{x \rightarrow 1^+} g(x)$ does not exists (B) $\lim_{x \rightarrow 1^-} g(x)$ does not exists
(C) $\lim_{x \rightarrow 1^+} g(x) = 0$ (D) $\lim_{x \rightarrow 1^-} g(x) = 0$

माना फलन f तथा g इस प्रकार परिभाषित हैं कि

$$f(x) = \lim_{h \rightarrow 0} \frac{\sin(x+3h) - 3\sin(x+2h) + 3\sin(x+h) - \sin x}{h(\tan(x+2h) - 2\tan(x+h) + \tan x)} \text{ तथा}$$

$$g(x) = \frac{1-x(1+|1-x|)}{|1-x|} f\left(\frac{1}{1-x}\right) \sin \frac{1}{1-x} \sec^3 \frac{1}{1-x}.$$

तो निम्न में से कौनसा/से सही है/हैं ?

(A) $\lim_{x \rightarrow l^+} g(x)$ विद्यमान नहीं है

(B) $\lim_{x \rightarrow l^-} g(x)$ विद्यमान नहीं है

(C) $\lim_{x \rightarrow l^+} g(x) = 0$

(D) $\lim_{x \rightarrow l^-} g(x) = 0$

Ans.

AD

Sol.

Numerator :

$$2 \cos\left(x + \frac{3h}{2}\right) \sin \frac{3h}{2} - 6 \cos\left(x + \frac{3h}{2}\right) \sin \frac{h}{2}$$

$$2 \cos\left(x + \frac{3h}{2}\right) \left[\sin \frac{3h}{2} - 6 \sin \frac{h}{2} \right]$$

Denominator :

$$\tan(x + 2h) - 2 \tan(x + h) + \tan x$$

$$\tan(x + 2h) - \tan(x + h) + \tan x - \tan(x + h)$$

$$\frac{\sin(x + 2h) \cos(x + h) - \sin(x + h) \cos(x + 2h)}{\cos(x + 2h) \cos(x + h)} - (\tan(x + h) - \tan x)$$

$$\frac{\sinh}{\cos(x + 2h) \cos(x + h)} - \frac{\sinh}{\cos x \cos(x + h)}$$

$$\frac{\sinh}{\cos(x + h)} \left[\frac{\cos x - \cos(x + 2h)}{\cos x \cos(x + 2h)} \right]$$

$$\frac{\sinh}{\cos(x + h)} \left[\frac{2 \sin(x + h) \sin(h)}{\cos x \cos(x + h)} \right]$$

$$f(x) = \lim_{h \rightarrow 0} \frac{2 \cos\left(x + \frac{3h}{2}\right) \left(\sin \frac{3h}{2} - 6 \sin \frac{h}{2} \right)}{2h \sin^2 h \sin(x + h)} \cos^2(x + h) \times \cos x$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{\cos\left(x + \frac{3h}{2}\right) \cos^2(x + h) \cos x}{\sin(x + h) h \times h^2} \left[\frac{3h}{2} - \frac{\left(\frac{3h}{2}\right)^3}{3!} + \dots - \frac{6h}{2} - 6 \frac{\left(\frac{h}{2}\right)^3}{3!} + \dots \right]$$

$$f(x) = \frac{\cos^4 x}{\sin x} \left[\frac{27}{8} \times \frac{1}{6} - 6 \times \frac{0}{8 \times 6} \right]$$

$$f(x) = \frac{\cos^4 x}{\sin x} \left(\frac{9}{16} - \frac{2}{16} \right)$$